



**RESOURCE GUIDE FOR
THE COURSE ON
PROJECT MANAGEMENT**

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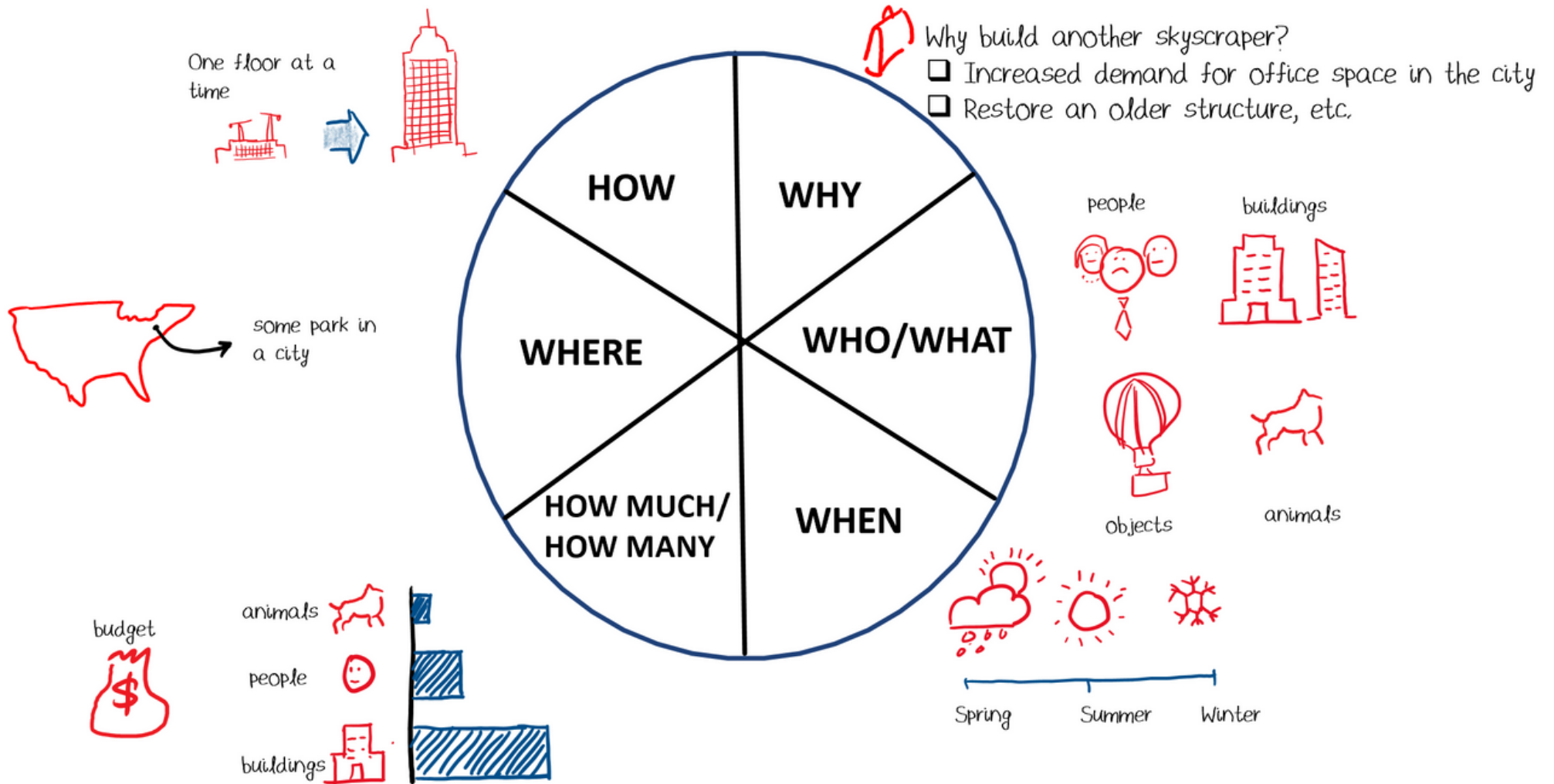
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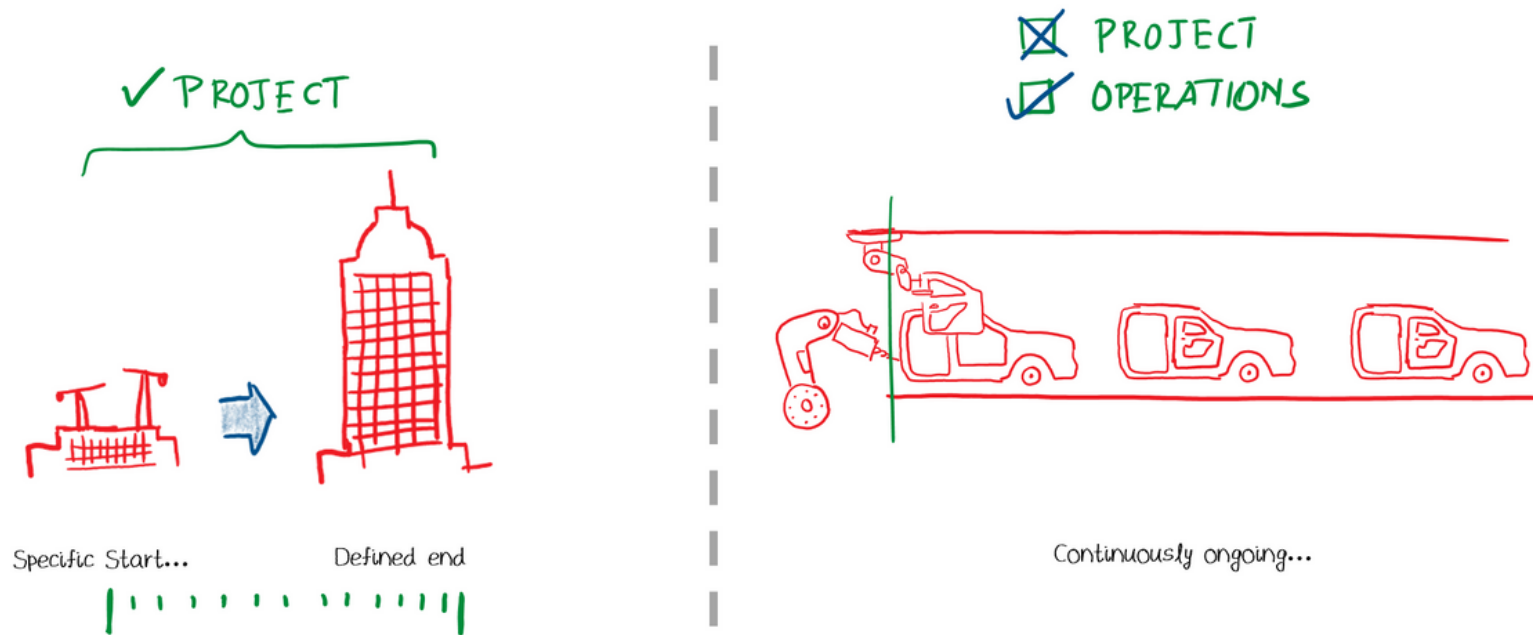
THE BIG PICTURE OF EVERY PROJECT



Six visual pathways can help us understand project management:

- The first pathway is the WHO and the what, which includes stakeholders and the project's product or service
- The second pathway is the WHEN – the dimension of time, which includes the project's schedule
- The third pathway is the HOW MUCH/ HOW MANY, which involves quantitatively determining the resources needed for the project
- The fourth pathway is WHERE, which looks at the spatial lines and hierarchical arrangements within organizations involved in the project
- The fifth pathway is HOW, which involves choosing an appropriate methodology or framework for the project
- The sixth pathway is WHY, which considers the business rationale for the project and whether it's worth the effort
- These pathways align with how we see things in the real world.
- Thinking project management in this way makes it intuitive to manage projects

WHAT IS A PROJECT



A project is a temporary endeavor undertaken to create a unique product, service or result.

- Projects are temporary work with a defined start and end, aimed at creating a unique product, service, or result that didn't exist before.
- Projects involve a coordinated set of activities with a defined plan to follow.
- Examples of projects include building a physical structure or creating a website from scratch, while ongoing activities without a defined start and end are considered operations.
- Car manufacturing is not a project, but creating a unique car design is.
- Projects can end prematurely due to lack of budget or support from key stakeholders.

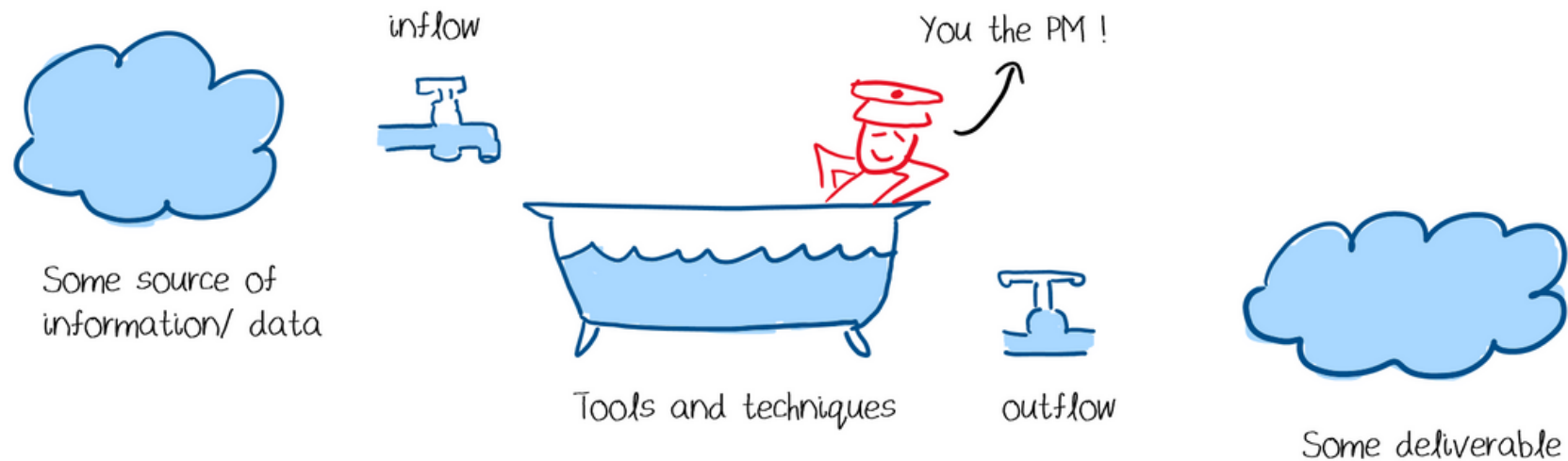
WHAT IS PROJECT MANAGEMENT



"Project Management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements", and a Project Manager is that individual assigned by the organization to skillfully lead the team to achieving the project objectives.

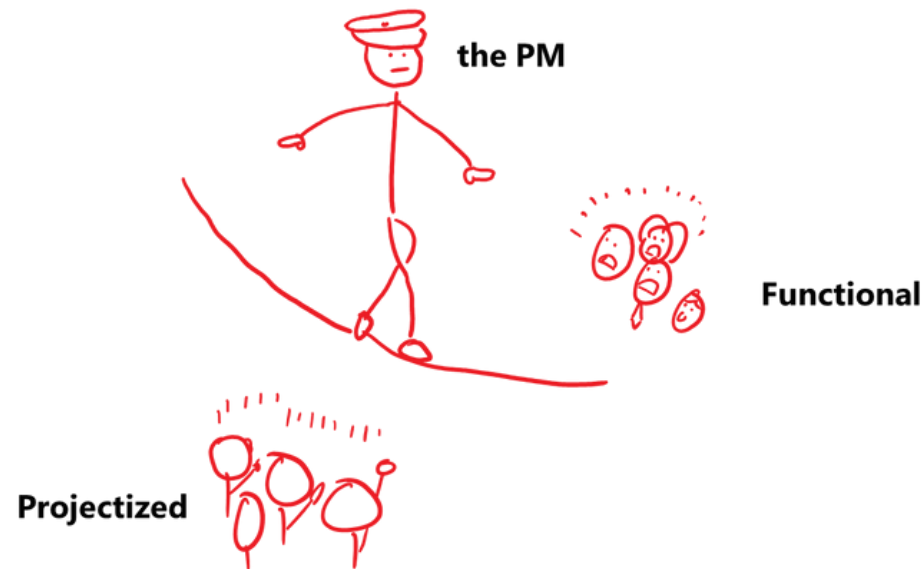
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- Project management guides us in coordinating activities for a project.
 - Project management applies knowledge, skills, tools, and techniques for structured methodology.
 - Without project management, we wouldn't know where to start or how to approach the work.
 - Project management is about efficiently juggling multiple activities and maintaining balance.
 - A project manager coordinates activities through project management methodologies and principles.

SYSTEMS THINKING



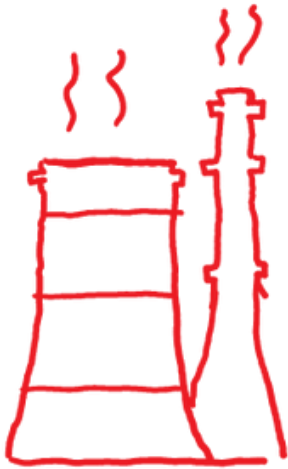
- ❑ Systems thinking is used to visualize all the inputs and outputs in project management processes.
- ❑ Inputs are fed into a process, where they are mixed together using various tools and techniques to produce outputs.
- ❑ The PM is the primary person facilitating this process.
- ❑ The outputs are interim project deliverables like project plans, schedules, budgets, and the actual product.
- ❑ This is a simplified representation of the systems thinking concept, borrowed from systems theory and modified for project management purposes.

MATRIXED ENVIRONMENT



- Understanding the organization's structure and hierarchy is important for project managers.
- There are mainly three types of organizational structures: functional, project-based, and matrix.
- Functional organizations are based on specialties and give authority to functional managers.
- Project-based organizations give authority to project managers and are focused on project delivery.
- Matrix structures share authority between functional and project managers.
- Balanced matrix environments are ideal but rare.
- Weak matrix environments give more authority to functional managers, while strong matrix environments give more authority to project managers.
- Project managers have varying levels of authority over resources and decision-making depending on the type of organizational structure.

EEF AND OPA



Enterprise Environmental Factors

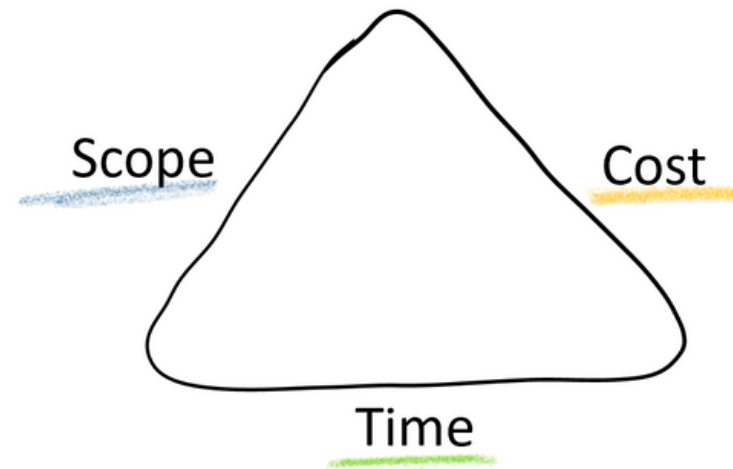
- Government regulations
- Industry standards
- Market conditions
- Political climate

Organizational Process Assets

- Internal policies and procedures
- Guidelines
- Templates
- Systems
- Best practices

-
- Enterprise Environmental Factors: Not everything in a project is under your control, and external factors may have an influence on the outcome.
 - These external factors are called Enterprise Environmental Factors, which may include government regulations, market conditions, and political climates.
 - Organizational Process Assets are an organization's internal policies, procedures, guidelines, templates, systems, and best practices that may have been collected from past experiences. These assets are important inputs to many project management processes.

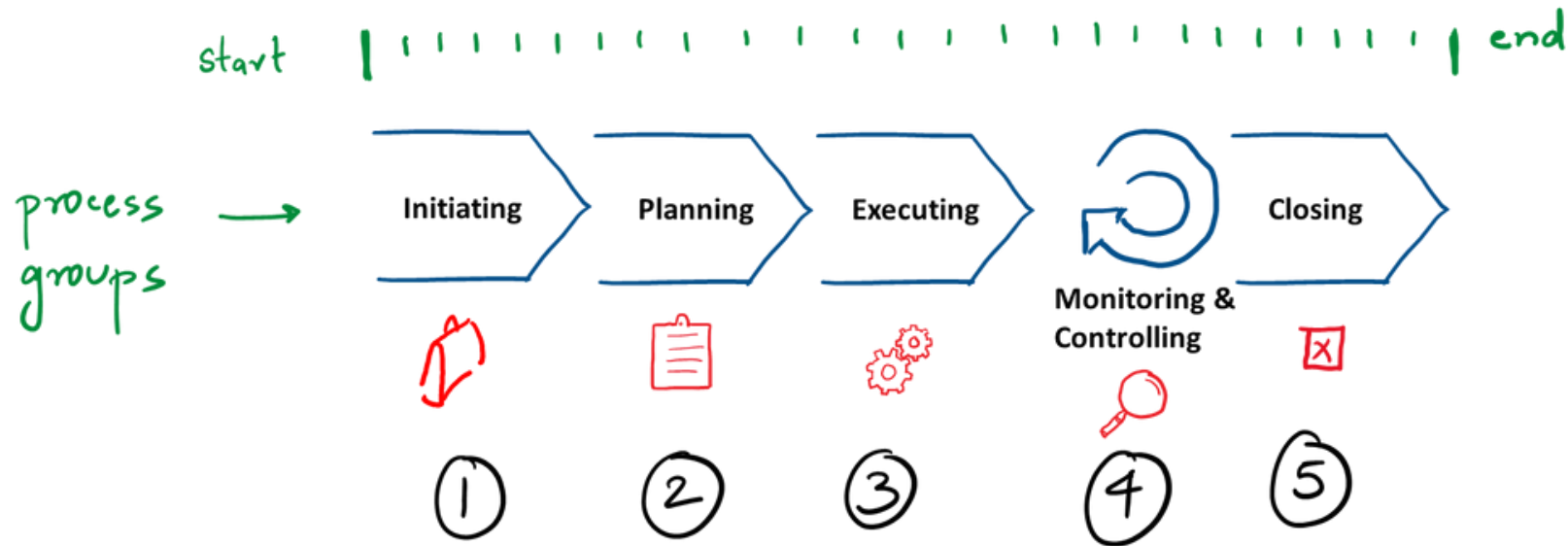
PROJECT CONSTRAINTS



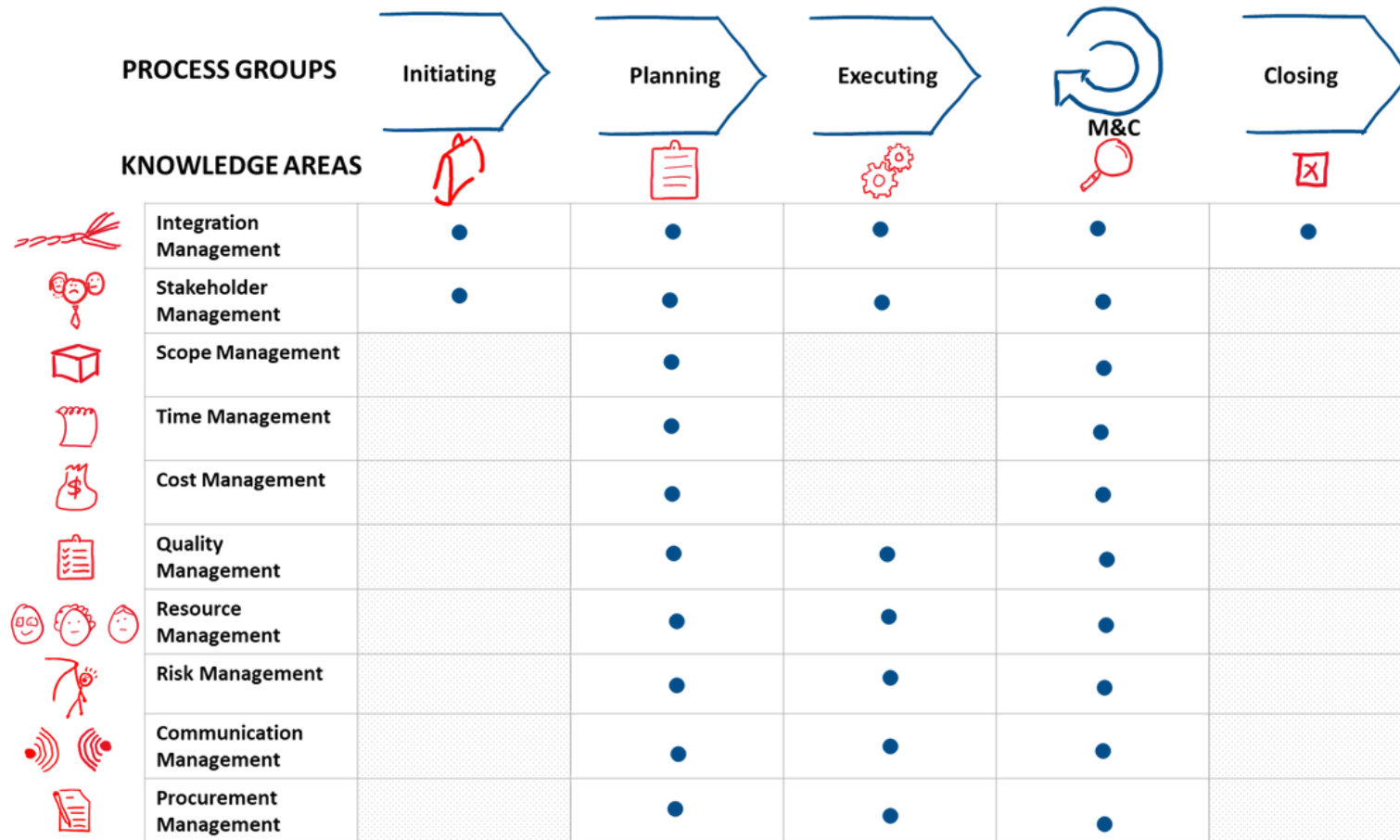
“Iron triangle” because when you alter one project constraint you will end up altering other project constraints

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- ❑ Project managers need to understand that there are limitations to the resources, time, and budget available for a project.
 - ❑ These limitations are called project constraints and they dictate the actions that can be taken by the project manager and team.
 - ❑ Constraints can be of different types and not just limited to scope, time, or cost.
 - ❑ The iron triangle is a concept that refers to the three constraints of scope, time, and budget that dictate the setup and execution of a project.

PROCESS GROUPS

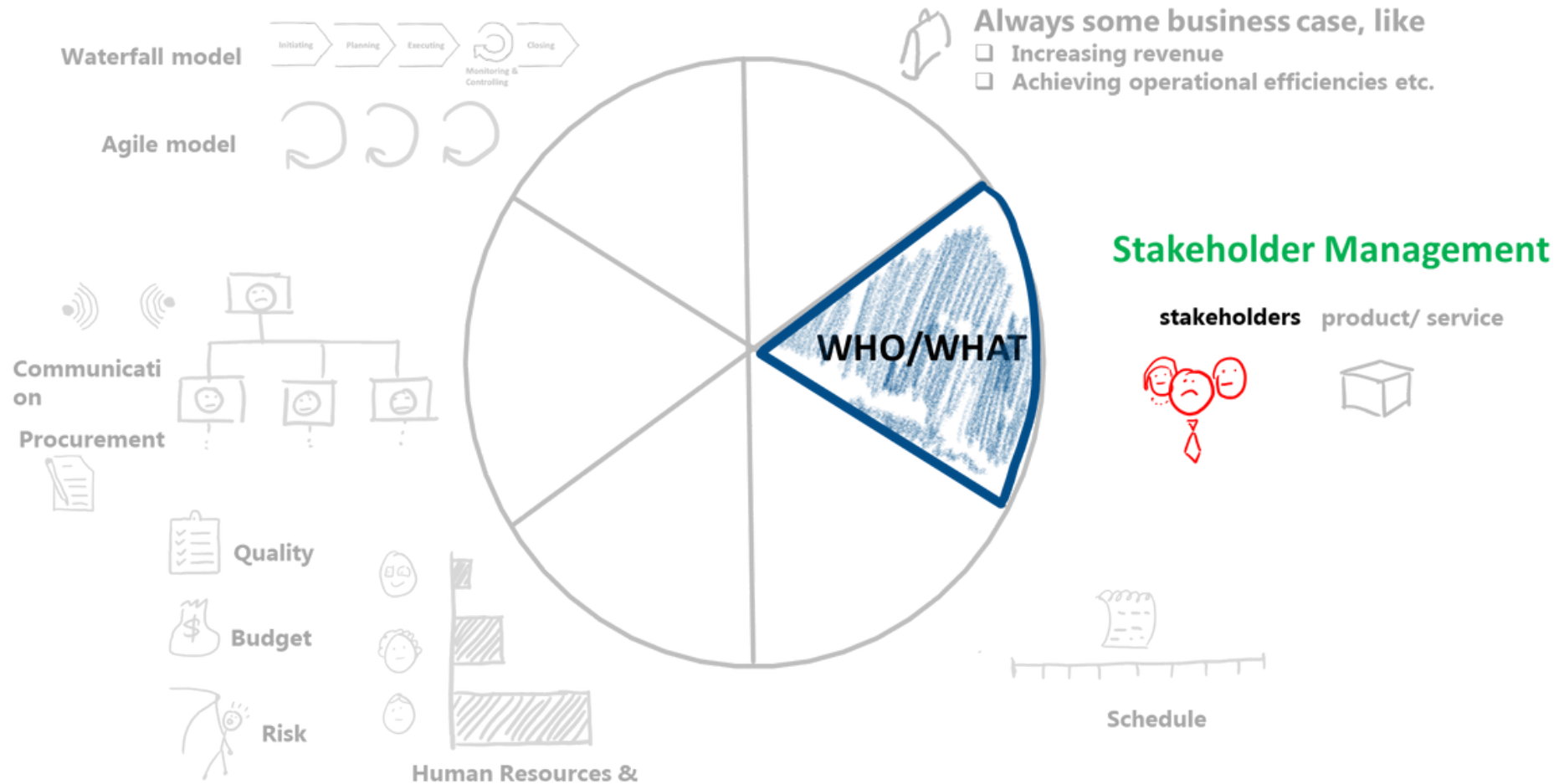


- Time is a crucial component in a project.
- Projects are divided into five main process groups: initiating, planning, executing, monitoring and controlling, and closing.
- The initiating process group involves putting everything needed to start the project and officially recognizing it.
- The planning phase is where most of the work is done in planning for the project's future, and the project management plans are created.
- The executing phase involves building the final product or service and coordinating with people and resources.
- The monitoring and controlling phase involves tracking progress, identifying areas of changes, and initiating corrective and preventive actions.
- The closing phase includes concluding activities and closing all project management processes and contractual obligations.



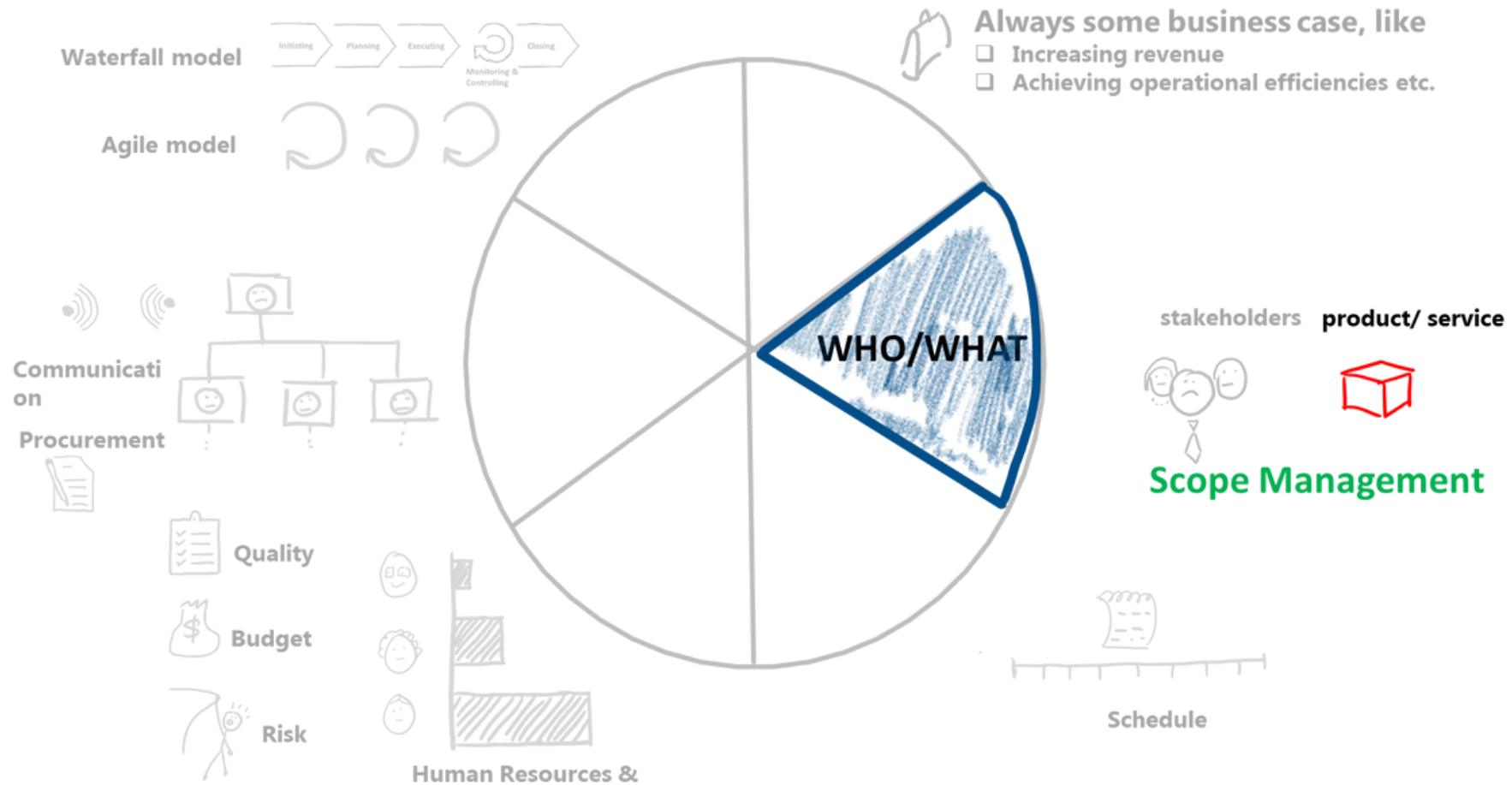
- Project management is divided into 10 knowledge areas and 5 process groups
- The blue dots represent PM Processes in those Knowledge Areas and Process Groups

KNOWLEDGE AREAS



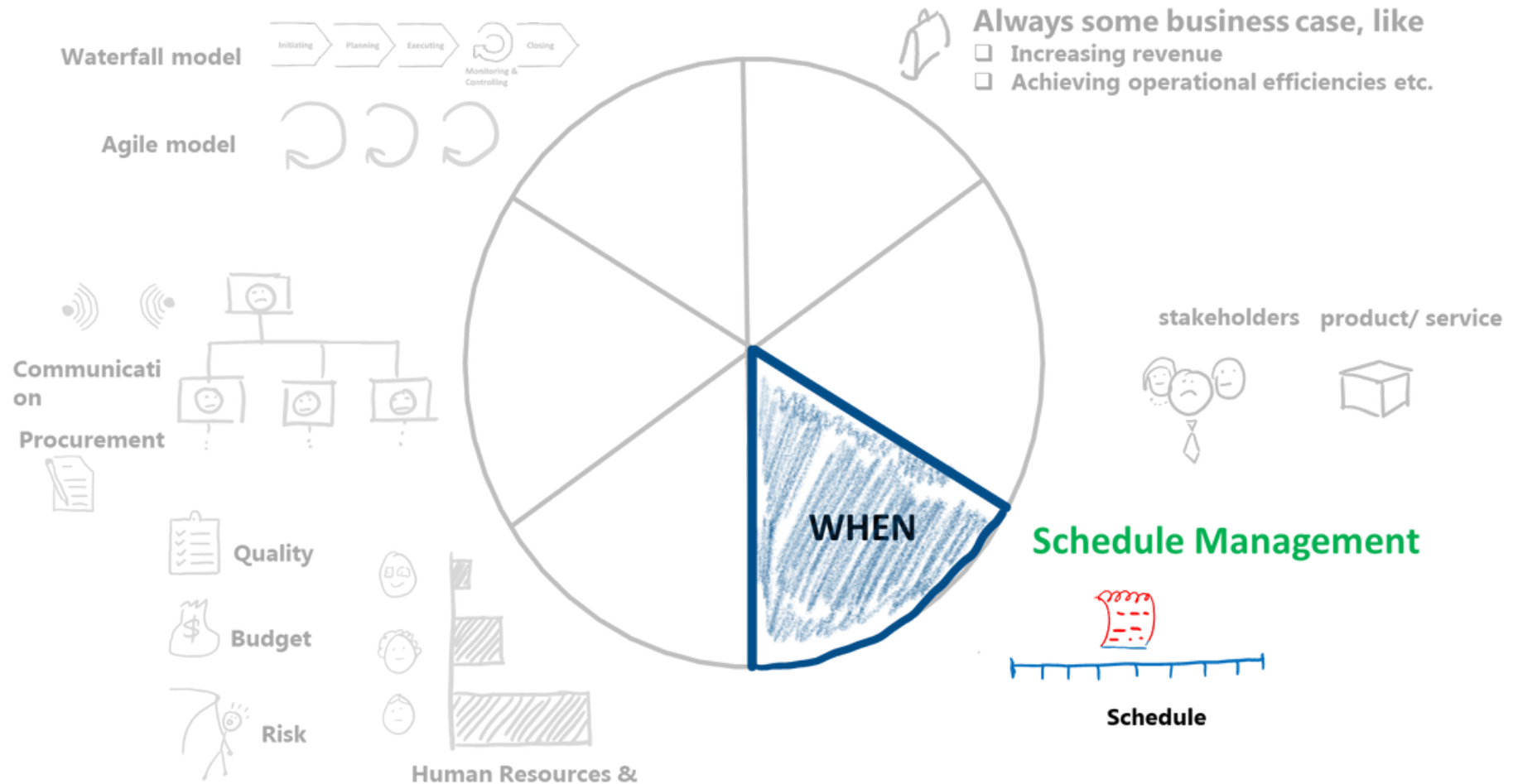
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- Stakeholder management is a knowledge area in project management that involves identifying, analyzing, and managing stakeholders who have an interest or influence in the project.
 - The main goal of stakeholder management is to ensure that stakeholders' needs and expectations are understood, addressed, and managed effectively to support the success of the project.
 - This includes identifying all the stakeholders and determining their level of involvement, assessing their impact on the project, and managing their expectations and concerns.

KNOWLEDGE AREAS



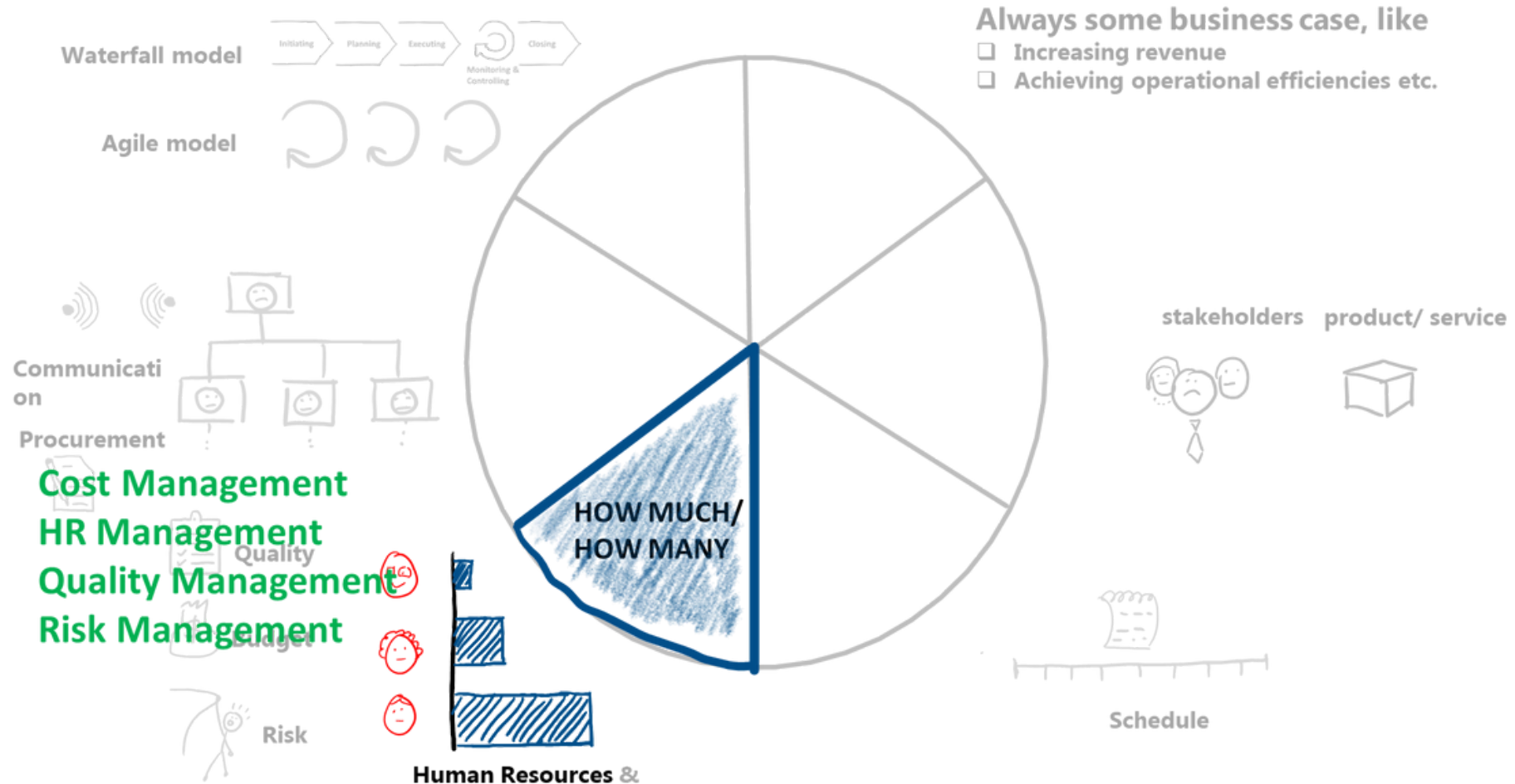
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- Scope management in project management refers to the processes and techniques used to define, validate, and control the work that needs to be done on a project.
 - It involves identifying all the project deliverables, requirements, and objectives, as well as establishing a plan for how to accomplish them.
 - The goal of scope management is to ensure that the project stays on track and that all project work is completed within the agreed-upon timeline and budget.

KNOWLEDGE AREAS



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- Schedule management is a knowledge area in project management that deals with the planning, development, monitoring, and control of project schedules.
 - It includes the processes required to ensure that the project is completed on time, taking into account all the necessary tasks and resources required to deliver the project objectives.
 - The main goal of schedule management is to create a schedule that is realistic, achievable, and meets the needs of all stakeholders.

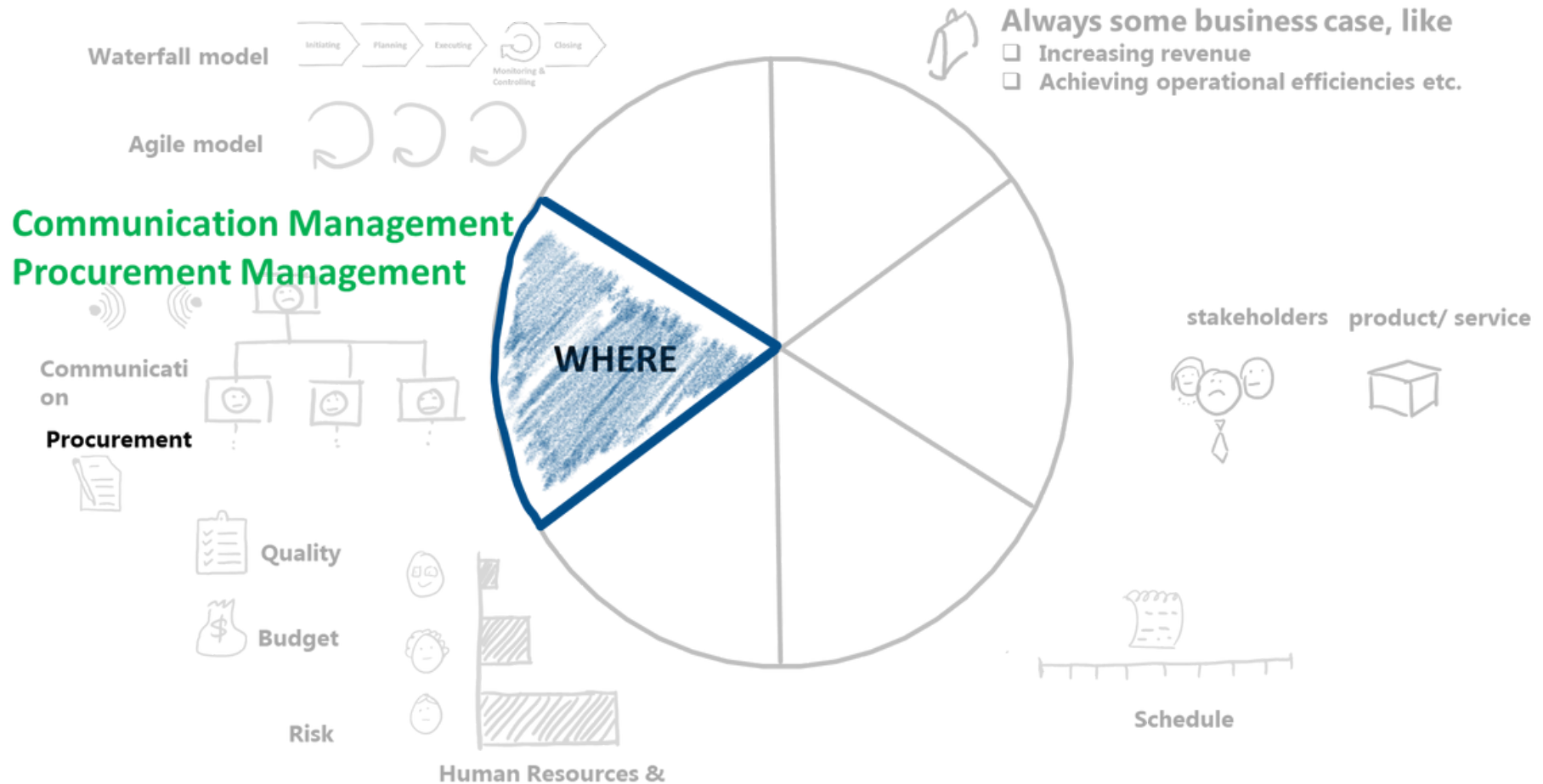
KNOWLEDGE AREAS



The next knowledge areas are quantitative and focus on measuring certain aspects of the project

- Cost management deals with estimating resources cost and establishing budgets to determine if the project continues or stops
- Resource management involves managing people working on the project, including conflicts and motivation, and ensuring their effective use
- Quality management ensures that the project meets the requirements and quality standards set out in the planning process
- Risk management is concerned with identifying and planning for potential risks and evaluating their probability of impact, including positive consequences that can improve project outcomes or efficiencies

KNOWLEDGE AREAS



-
- The Communications Management and Procurement Management are two knowledge areas we will discuss in the WHERE portion.
 - Communication Management processes are related to general communication among stakeholders and the project team, and the project manager's communication skills are essential to coordinating and delivering the project.
 - Processes in Communication Management ensure that project information, including plans, assessments, and meetings, are collected, documented, and archived properly.
 - Procurement Management processes are involved in purchasing goods or services from external vendors or contractors, and discussions assume the perspective of a buyer.

KNOWLEDGE AREAS

PROCESS GROUPS		Initiating	Planning	Executing	 M&C	Closing
KNOWLEDGE AREAS						
	Integration Management	Develop Project Charter Process	•	•	•	•
	Stakeholder Management	•	•	•	•	
	Scope Management		•		•	
	Time Management		•		•	
	Cost Management		•		•	
	Quality Management		•	•	•	
	Resource Management		•	•	•	
	Risk Management		•	•	•	
	Communication Management		•	•	•	
	Procurement Management		•	•	•	

Develop Project Charter Process is in the Initiating Process group and in the Integration Management Knowledge Area

WHY DO THIS PROJECT

Business case



There always must be a business case to start a project, like...

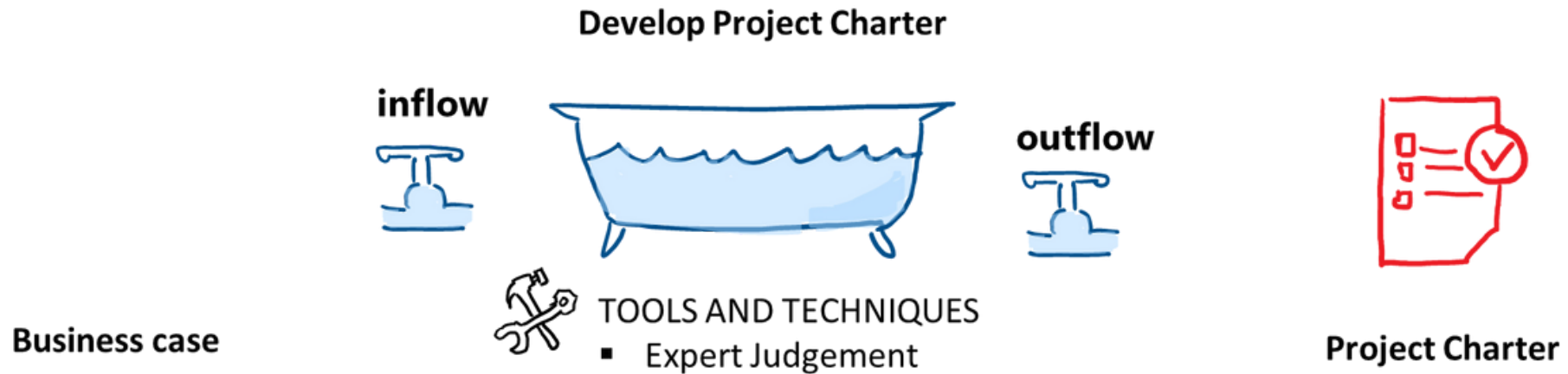
- ☐ Increasing revenue
- ☐ Achieving operational efficiencies etc.

- ① The business case establishes the fundamental rationale behind starting the project
- ② It talks about what your company is trying to get out of the project, whether the investment in the project is worth the effort, and how much money can they make or save from doing it
- ③ Describes the cost-benefit analysis

A business case typically outlines the following:

- Problem statement: A clear and concise explanation of the problem or opportunity that the project aims to address.
- Objectives: The specific goals and objectives of the project, including how they align with the overall strategic objectives of the organization.
- Benefits: The expected benefits of the project, both tangible and intangible, and how they will contribute to the organization's bottom line.
- Costs: An estimate of the total cost of the project, including direct and indirect costs, as well as the expected return on investment.
- Risks: Identification and assessment of potential risks associated with the project, along with mitigation strategies.
- Alternatives: An evaluation of alternative solutions to the problem or opportunity, along with a comparison of their costs and benefits.
- Stakeholders: Identification of all stakeholders impacted by the project, along with their interests, needs, and concerns.

WHY DO THIS PROJECT



PROJECT CHARTER

- ① The project charter is the official, written acknowledgment and recognition that a project exists.
- ② It gives the project manager the authority to assign organizational resources to the project.
- ③ The charter documents the business need or demand that the project was initiated to address, and it includes a description of the product, service, or result of the project.

-
- ❑ The develop project charter process converts the business case into an official document that authorizes the project manager and assigns resources to the project.
 - ❑ The project charter is signed by a senior executive and gives the project manager full authority to start the project, obtain resources, and access project funds.
 - ❑ The expert judgment technique is used to review and agree on the sections of the project charter.

WHY DO THIS PROJECT

PROCESS GROUPS		Initiating	Planning	Executing	M&C	
KNOWLEDGE AREAS						
	Integration Management	•	•	•	•	•
	Stakeholder Management	Identify Stakeholders	•	•	•	
	Scope Management		•		•	
	Time Management		•		•	
	Cost Management		•		•	
	Quality Management		•	•	•	
	Resource Management		•	•	•	
	Risk Management		•	•	•	
	Communication Management		•	•	•	
	Procurement Management		•	•		

Identify Stakeholders Process is in the Initiating Process group and in the Stakeholder Management Knowledge Area

WHO IS ON THE PROJECT



**Project Sponsors/
Customer**



**Functional
Managers**



Project Manager

STAKEHOLDERS

- Any person or group who have a vested interest in the outcome of your project, and usually have something to gain or lose as a result

-
- Stakeholders can be internal or external to the organization and can include individuals, groups, or organizations.
 - Internal stakeholders may include project sponsors, project managers, team members, and other employees of the organization. External stakeholders may include customers, suppliers, government agencies, and community groups.
 - Effective stakeholder management is essential for project success, as it helps to ensure that stakeholders are engaged, informed, and supportive of the project.

WHO IS ON THE PROJECT



inflow



outflow

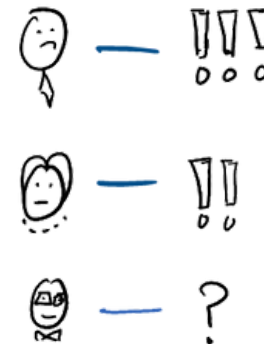


TOOLS AND TECHNIQUES

- Expert Judgement
- Meetings
- Stakeholder analysis and mapping

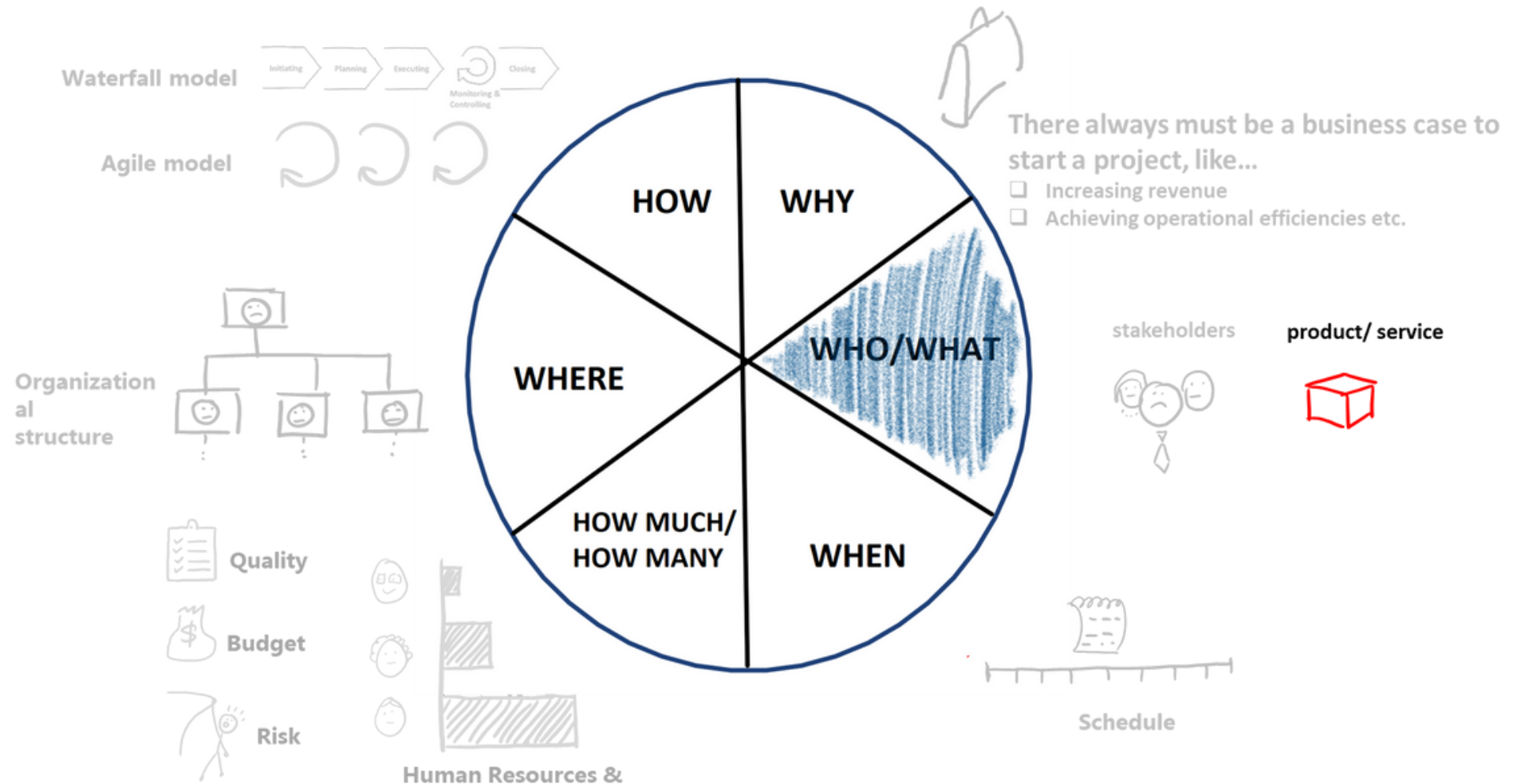


Stakeholder Register



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- It is important for the project manager to identify all important people on the project, known as stakeholders.
 - The process of identifying stakeholders occurs during the initiating phase and is not the process for identifying project team members or resources.
 - Key stakeholders include project sponsors, customers, and functional managers in a matrix environment.
 - The project manager assumes total responsibility for the project and is authorized by the project charter as a key stakeholder.
 - It is the responsibility of the project manager to establish ground rules, communication channels, and keep track of changing priorities of stakeholders through a stakeholder register.
 - Stakeholder analysis is used to understand their specific needs and expectations, and it is important to know what motivates them and their thresholds for support on the project.
 - The stakeholder register should be constantly updated to keep track of changing priorities.
 - Stakeholders may have varying levels of influence and interest in the project, and their needs and expectations must be considered throughout the project lifecycle.

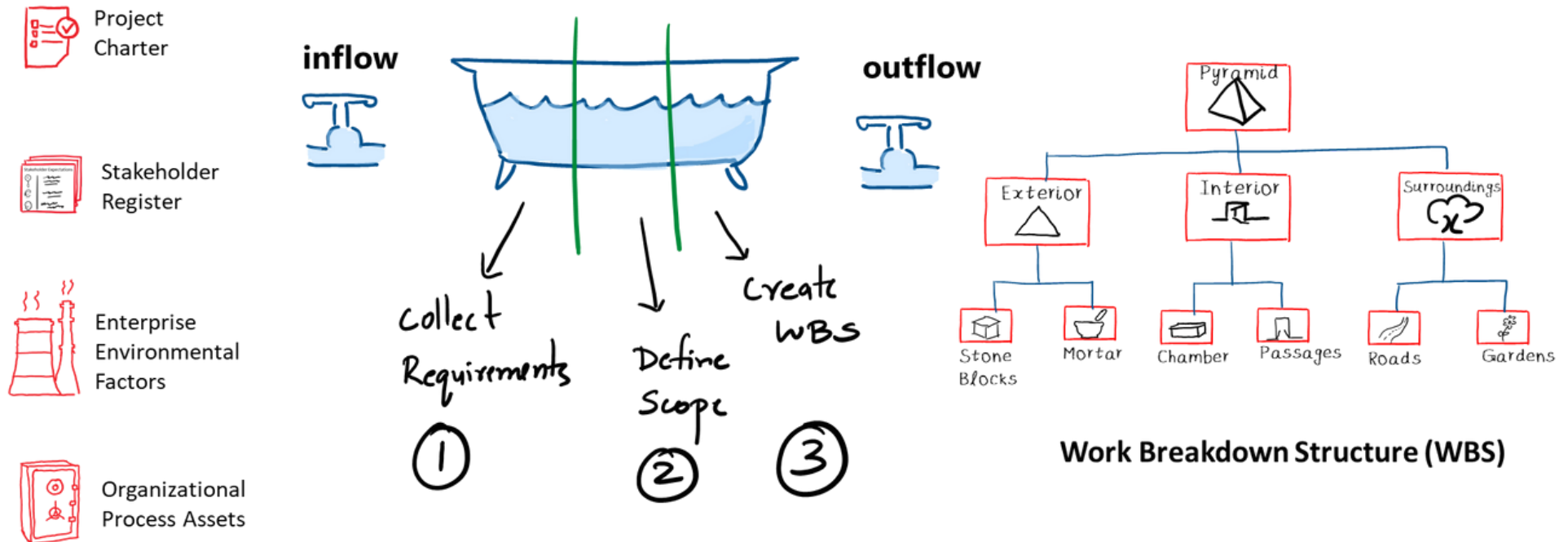
WHAT IS THE PROJECT SCOPE



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- ❑ Project planning ensures goals can be achieved within time and budget constraints
 - ❑ Scope management is important for defining project scope
 - ❑ Scope includes product scope (specifications of the product/service) and project scope (work needed to build the product/service)
 - ❑ Product scope defines the characteristics and qualities of the final deliverable. It encompasses features, functions, and requirements of the product, such as size, shape, design, performance, and any other specific requirements that are essential to meet the product's goals and objectives.
 - ❑ Project scope, on the other hand, covers all the work that is required to complete the project. It includes project management activities, such as planning, organizing, directing, and controlling the project, as well as tasks that are required to complete the product.
 - ❑ Project scope considers the project's goals, objectives, timelines, budgets, and resources that are required to complete the project successfully.

WHAT IS THE PROJECT SCOPE

SCOPE MANAGEMENT



- Work breakdown structure (WBS) is crucial output of scope management
- The work breakdown structure (WBS) is a visual tool that captures what is in the scope statement.
- WBS is created through three processes: collect requirements, define scope, and create WBS
- Inputs for creating WBS include project charter and stakeholder register

WHAT IS THE PROJECT SCOPE



Project
Charter

inflow



Stakeholder
Register



Enterprise
Environmental
Factors



Organizational
Process Assets



Requirements
Documentation

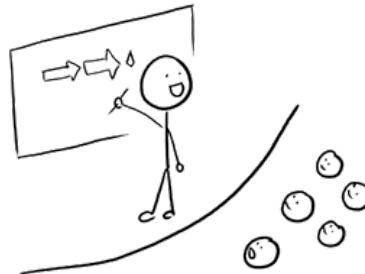


TOOLS AND TECHNIQUES

- Workshops
- Interviews
- Prototypes

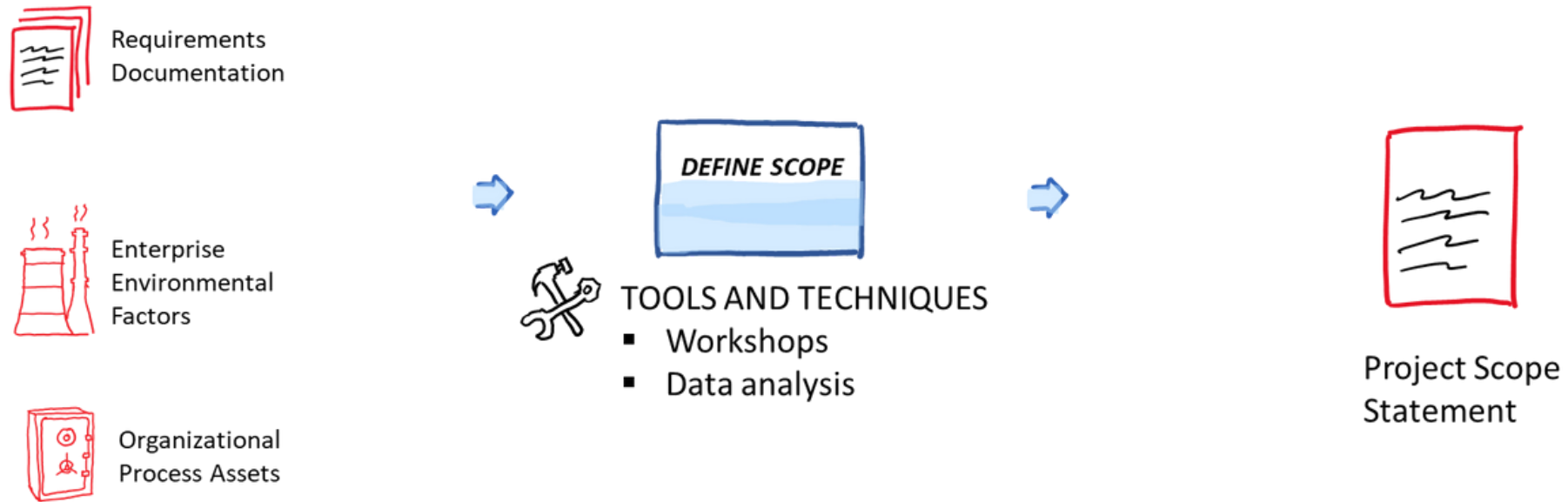
Source	Req.	Test
	~	✓
	~	x
	~	✓

Requirements Traceability Matrix



-
- ❑ Collect requirements is the first process in the scope management knowledge area.
 - ❑ Requirements are the characteristics of individual components of the product or service that become the scope of the project.
 - ❑ Requirements are from the perspective of the end user or customer who will ultimately be using the product or service.
 - ❑ Tools and techniques used in this process include workshops and interviews with stakeholders.
 - ❑ The output from this process is the requirements documentation.
 - ❑ The aim is to collect as many requirements as possible to satisfy stakeholders' expectations and needs.
 - ❑ It is important to trace back where individual requirements came from, which is captured in the requirements traceability matrix.

WHAT IS THE PROJECT SCOPE

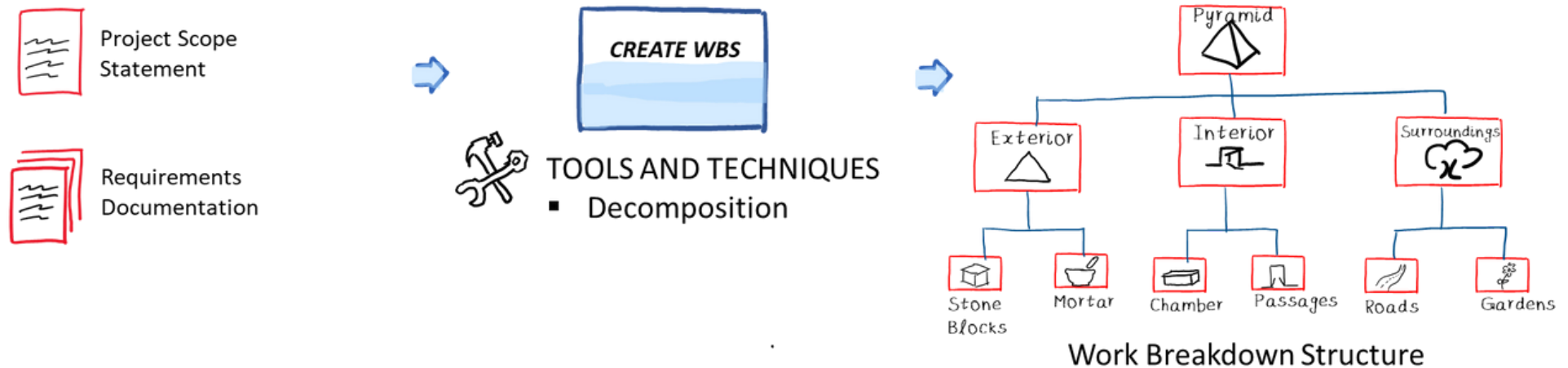


PROJECT SCOPE STATEMENT

- The purpose is to document exactly what is going to be delivered as part of the project, nothing more, nothing less
- It is an agreement between the project team and the customer that states precisely what the work of the project will produce

-
- ❑ The next step after collecting requirements is creating the project scope statement.
 - ❑ The project scope statement captures exactly what is and what is not in the scope of the project.
 - ❑ It includes acceptance criteria and assumptions made on the project.
 - ❑ The statement of work is sometimes closely linked to the project scope statement.
 - ❑ The project scope statement is created using workshops and data analysis.
 - ❑ It serves as an agreement between the project team and the customer, documenting exactly what will be delivered as part of the project.

WHAT IS THE PROJECT SCOPE

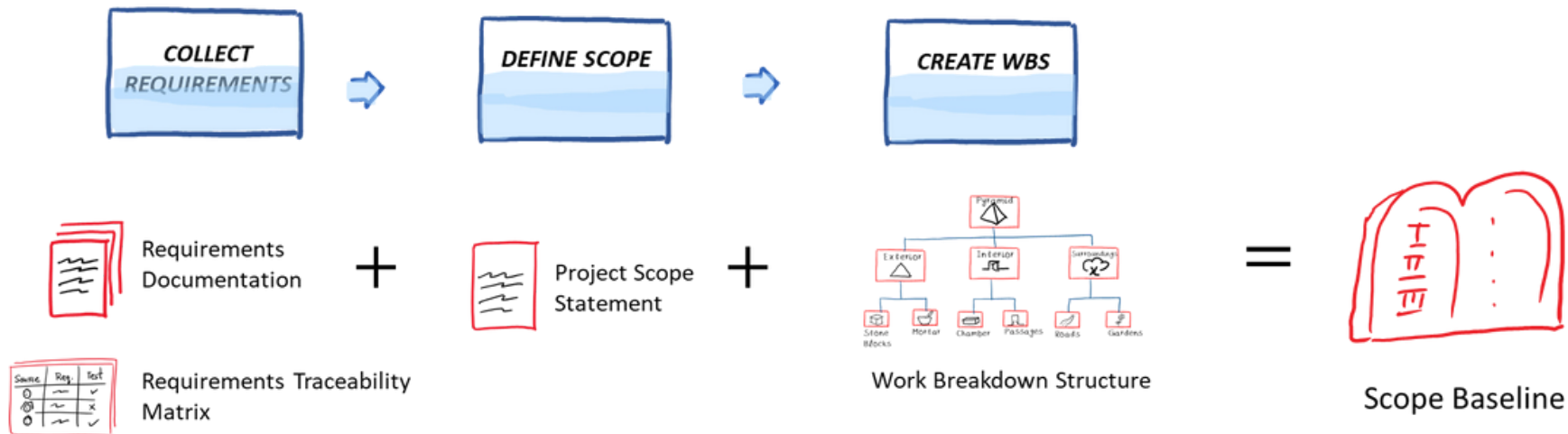


WORK BREAKDOWN STRUCTURE(WBS) is defined as the deliverable -oriented hierarchical decomposition of the work to be executed by the project team to accomplish all project objectives.

-
- ❑ A product or service is composed of individual components or parts.
 - ❑ Decomposition is a technique used to divide a large project scope into manageable components of work.
 - ❑ The Work Breakdown Structure (WBS) is a hierarchical decomposition of the work to be executed by the project team to accomplish project objectives.
 - ❑ The WBS helps to accurately estimate the costs, resources, and timelines for the entire project.
 - ❑ The WBS Dictionary contains additional information about the work, including descriptions, references, and contractual agreements.
 - ❑ Work packages are the smaller components of work that are more manageable and can be accurately estimated for their cost, resources, and timelines.

WHAT IS THE PROJECT SCOPE

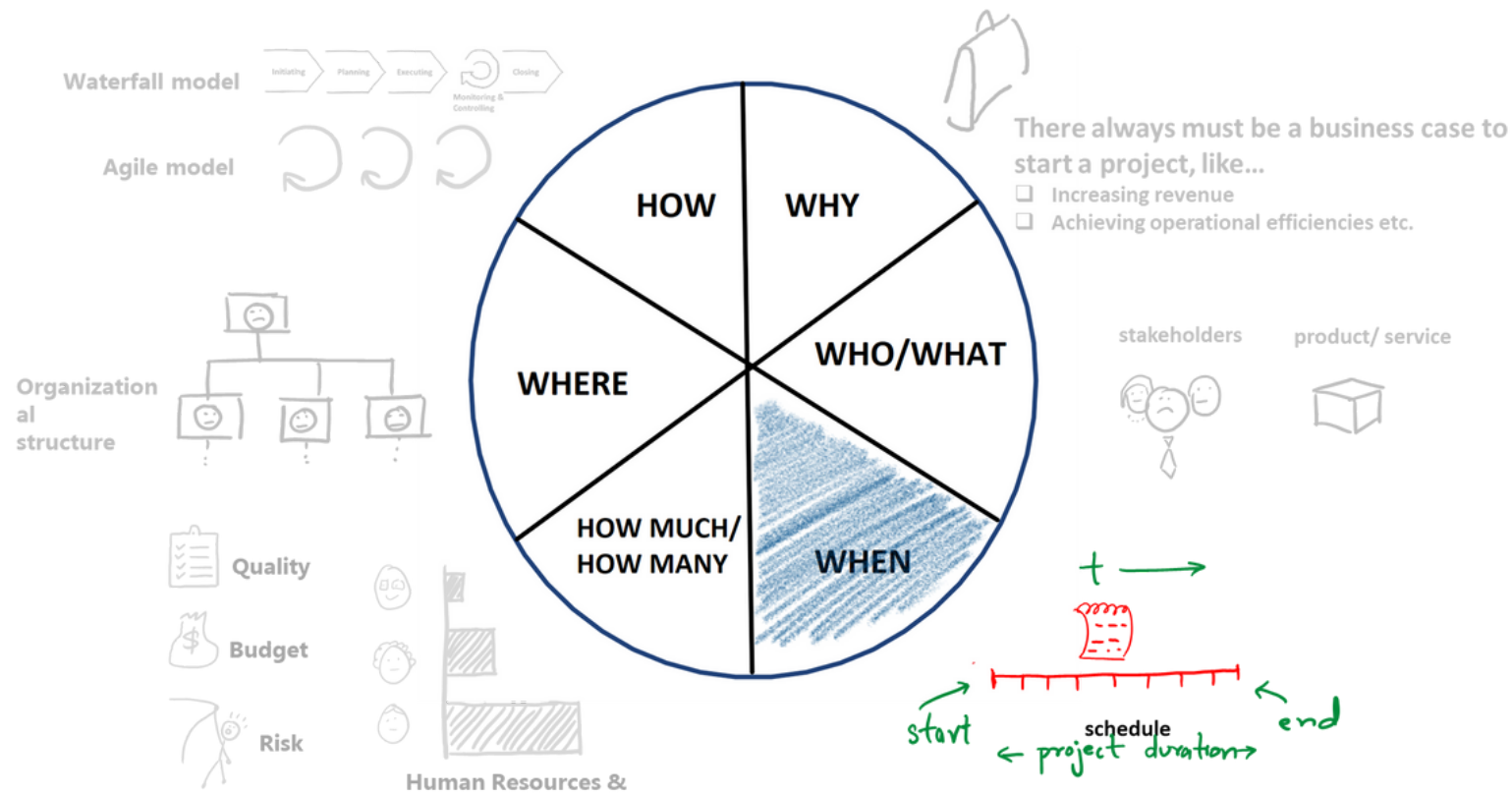
SCOPE BASELINE



- Scope baseline is the initial agreed-upon scope between the parties involved
- It is used as a benchmark to refer back during later phases to compare to in case of changes or variations from it

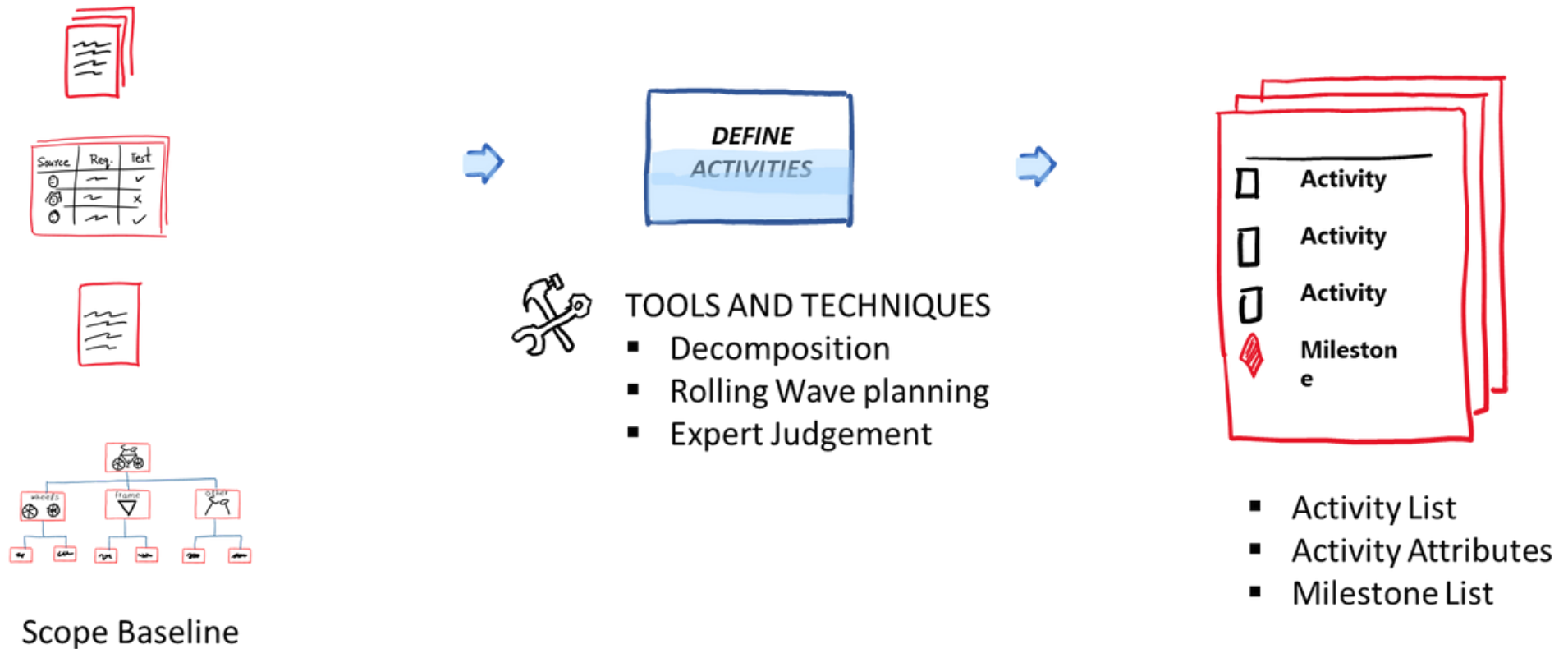
-
- ❑ Project managers often receive requests to add new scope or elements to the project
 - ❑ The decision to add or not to add is up to the project manager with approval from key stakeholders
 - ❑ The scope baseline is a collective sum of all the documents obtained from previous scope management processes
 - ❑ The scope baseline becomes the initial agreed-upon scope and is used as a benchmark for later phases of the project
 - ❑ Scope creep is any unapproved scope additions and should be rejected unless it goes through project change control procedure
 - ❑ Gold plating is adding features that do not bring significant improvement to the final product and should not be encouraged
 - ❑ The work breakdown structure is the most important output of the scope management knowledge area

WHAT IS THE PROJECT SCOPE



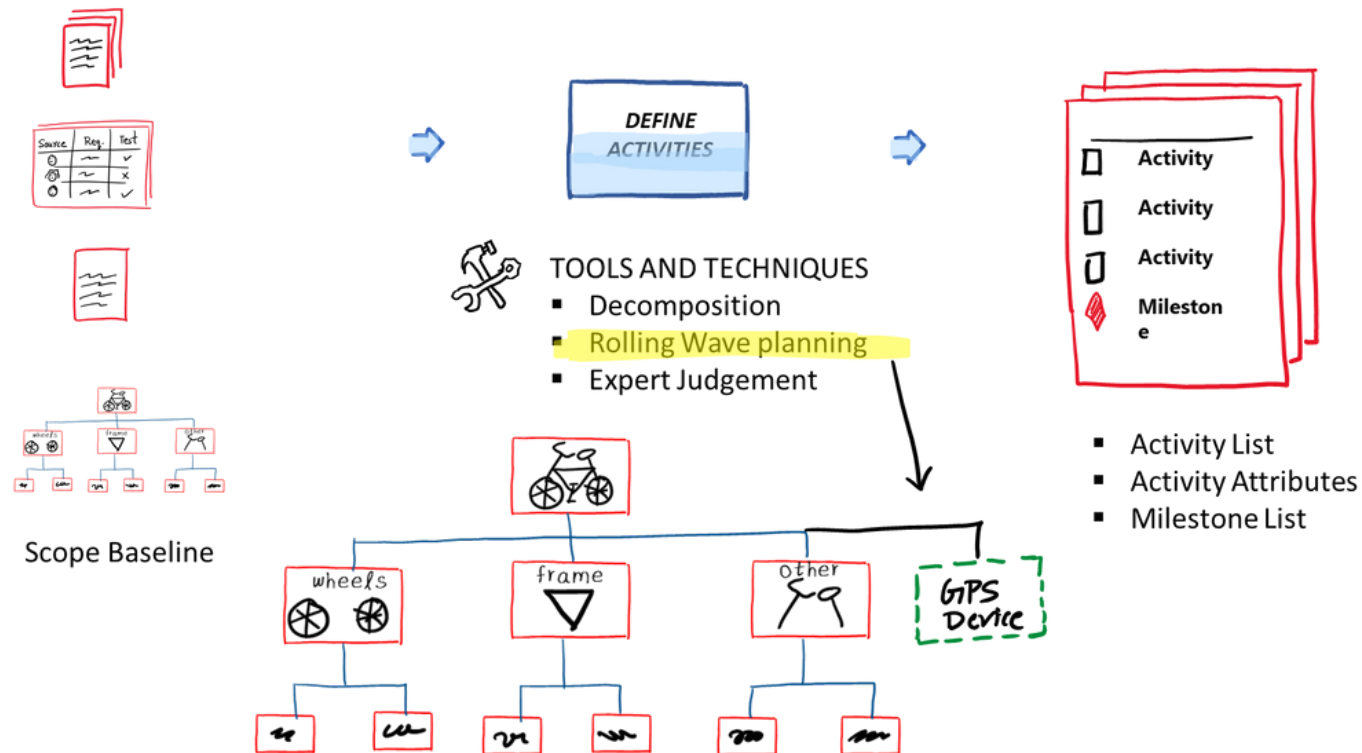
- Project management involves managing time as a key dimension
- Time is a limited resource and projects are temporary in nature
- Schedule management is a knowledge area that focuses on determining the project schedule and setting important milestones and deadlines

WHEN ARE THINGS TO BE DONE



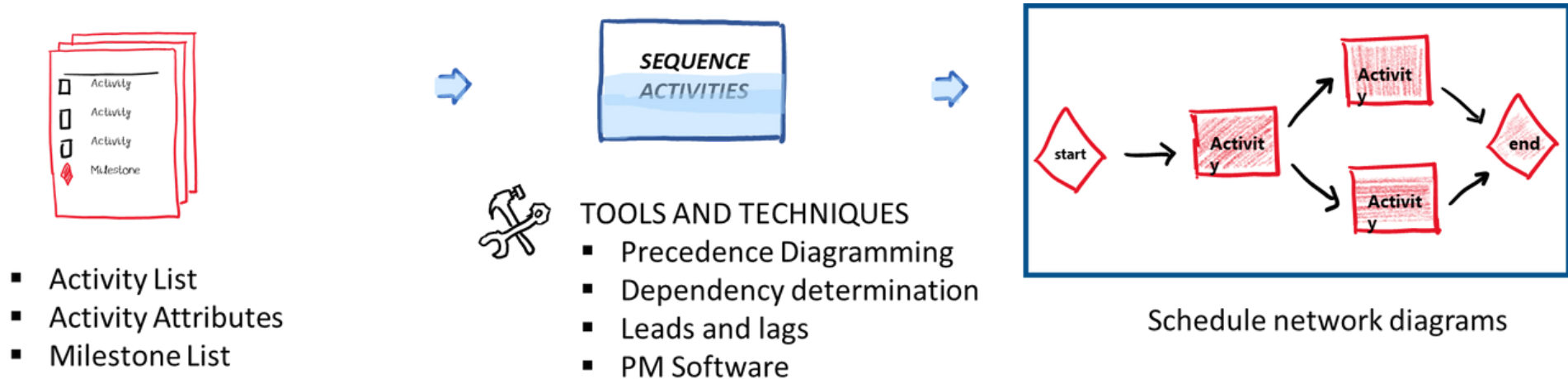
- Work packages are decomposed into activities, which are the tasks needed to complete the work packages
- Activity attributes document the details of activities, and the output of this process is a baseline

WHEN ARE THINGS TO BE DONE



- Rolling Wave Planning is a technique used in project management to elaborate on the details of the project plan gradually, as more information becomes available.
- It involves decomposing the project into work packages and identifying the activities necessary to complete each work package, but deferring the planning of activities until the work package is about to be executed.
- This allows for a more flexible approach to planning and allows for changes to be made as the project progresses.

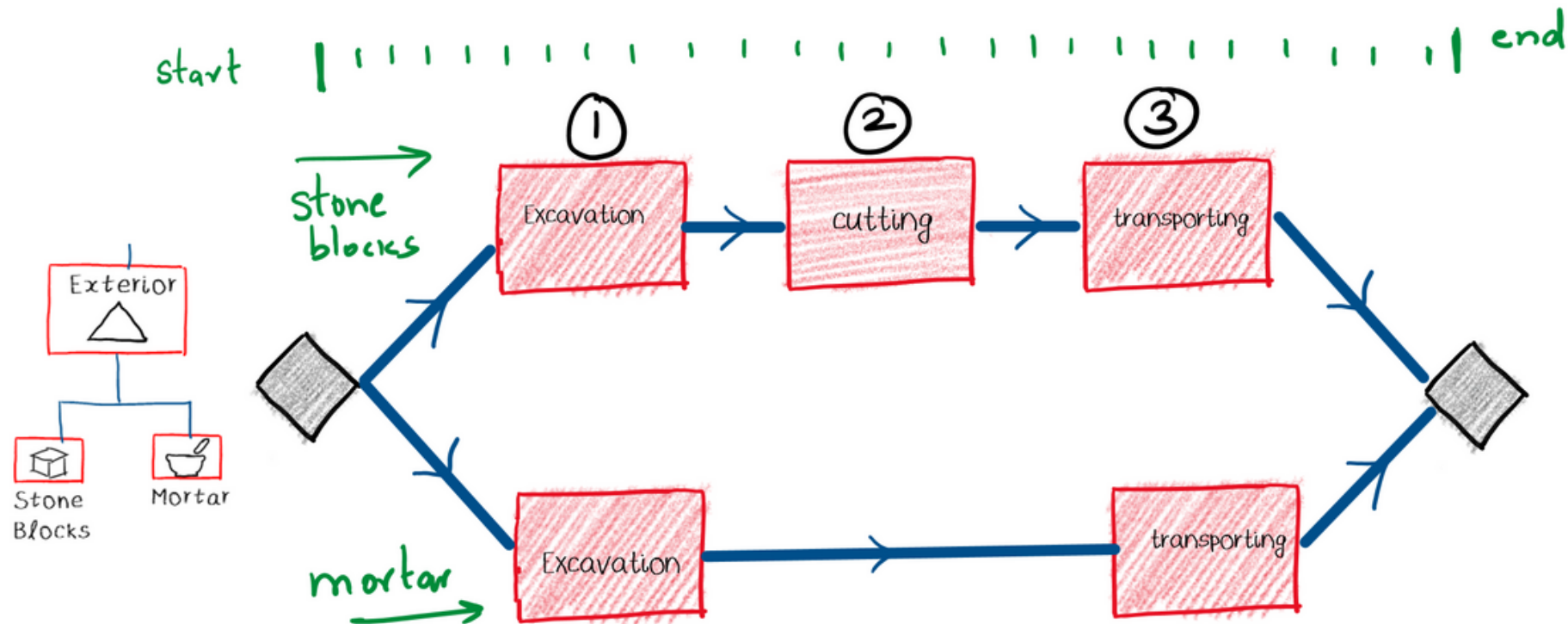
WHEN ARE THINGS TO BE DONE



- Sequencing of activities is the next step after identifying all the necessary activities and work packages in project management.
- It involves arranging the activities on a timeline and identifying the dependencies that exist between them.
- The Procedure Diagramming Method (PDM) is a tool used to determine dependencies and create a schedule network diagram.

WHEN ARE THINGS TO BE DONE

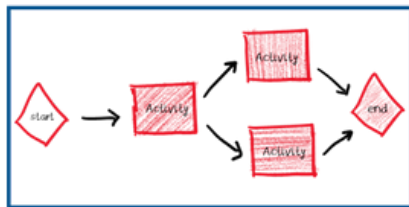
SEQUENCE ACTIVITIES



Schedule network diagram

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- The output from the Sequence Activities process is a schedule network diagram that is generated using the PDM, dependency determination, leads, lags, and project management software.
 - The sequencing of activities takes care of the relationships between these activities and helps to create a realistic schedule based on achievable facts.
 - The sequencing of activities under each work package is done through the PDM, and parallel activities that are independent of each other are shown to be happening concurrently.

WHEN ARE THINGS TO BE DONE

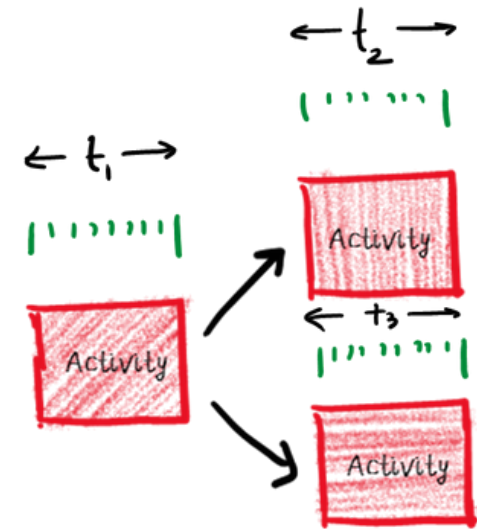


Schedule network diagrams



TOOLS AND TECHNIQUES

- Estimation methods

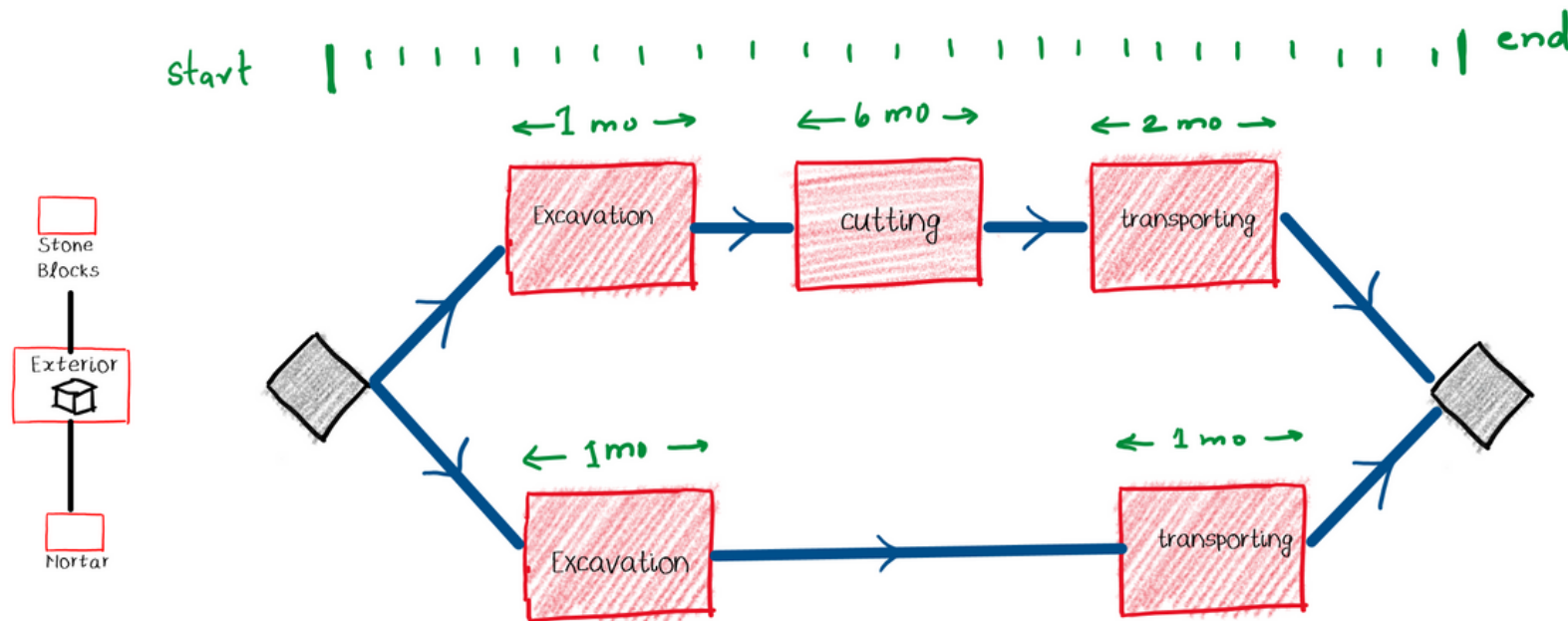


Activity duration estimates

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- Estimating durations accurately is the next step after sequencing activities.
 - The process estimates the time required for each activity to complete based on the previous prerequisite activities. It's important to know the precursor activities in order to estimate accurately.
 - Various estimation methods can be used, including current organizational process assets or knowledge gathered through internal or external means.
 - The output is the activity duration estimates, which will be used to create the final project schedule.

WHEN ARE THINGS TO BE DONE

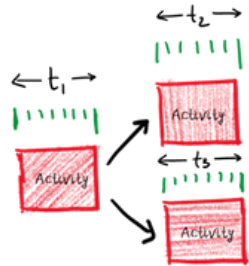
ESTIMATING ACTIVITY DURATIONS



Schedule network diagram

- Activity duration estimates are associated with each activity in the schedule network diagram.
- For example, in the pyramid project, excavation of stones takes one month, cutting takes six months, and transporting takes two months. Excavation and transporting for mortar take one month each.

WHEN ARE THINGS TO BE DONE



Activity duration estimates

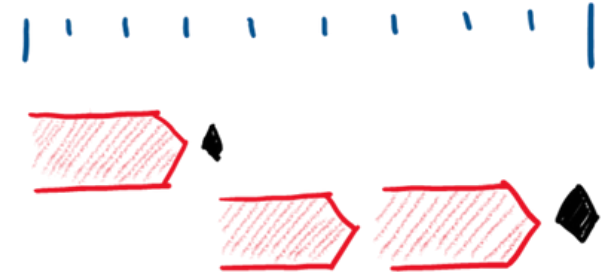


- Activity List
- Activity Attributes
- Milestone List



TOOLS AND TECHNIQUES

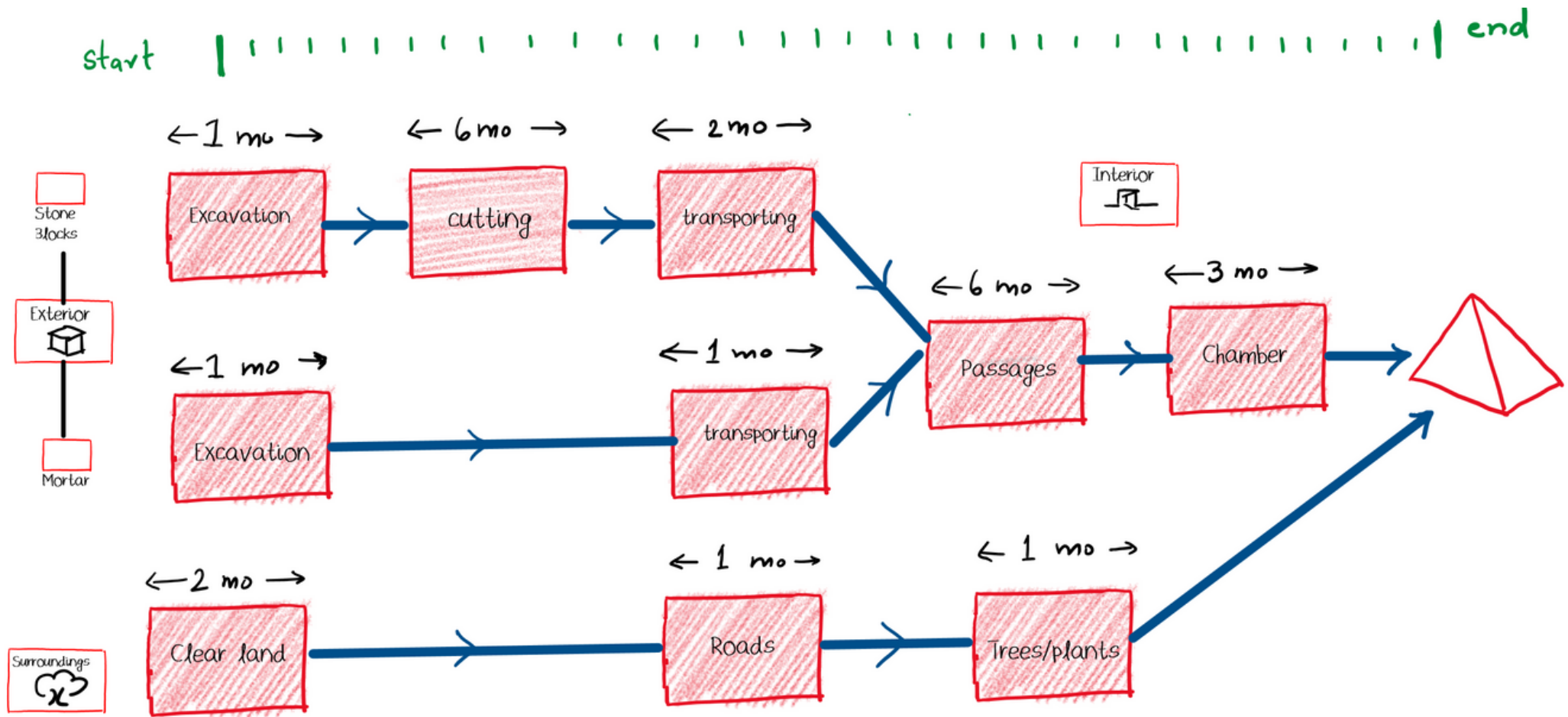
- Schedule Network Analysis
- Critical Path Method
- Leads and lags
- Schedule compression, etc.



Project Schedule

-
- ❑ Develop Schedule is the final step in project planning.
 - ❑ It produces the project schedule using activity duration estimates and the list of activities and milestones.
 - ❑ Various tools and techniques are used to create the project schedule.
 - ❑ The Critical Path method is used to determine the shortest amount of time to finish the project.
 - ❑ The Critical Path is the longest chain of tasks or activities in the schedule network diagrams, upon whose completion the project can be deemed completed.

WHEN ARE THINGS TO BE DONE

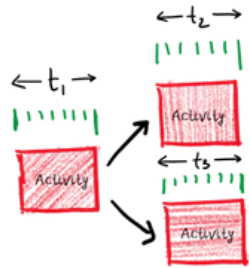


Schedule network diagram

In the example project pyramid:

- Three main paths in the diagram are: preparing stone blocks, preparing mortar, and preparing surroundings
- Critical path method determines the longest chain of tasks, which is the path for preparing stone blocks, cutting them, transporting, and building inside passages and final chamber, taking 18 months
- Activities on the critical path cannot be delayed and have zero float
- Other activities have float and can be delayed without jeopardizing the project schedule
- Final project schedule includes key milestones, including the date for final product delivery.

WHEN ARE THINGS TO BE DONE



Activity duration estimates

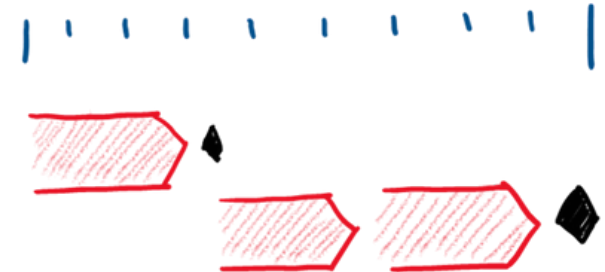


- Activity List
- Activity Attributes
- Milestone List



TOOLS AND TECHNIQUES

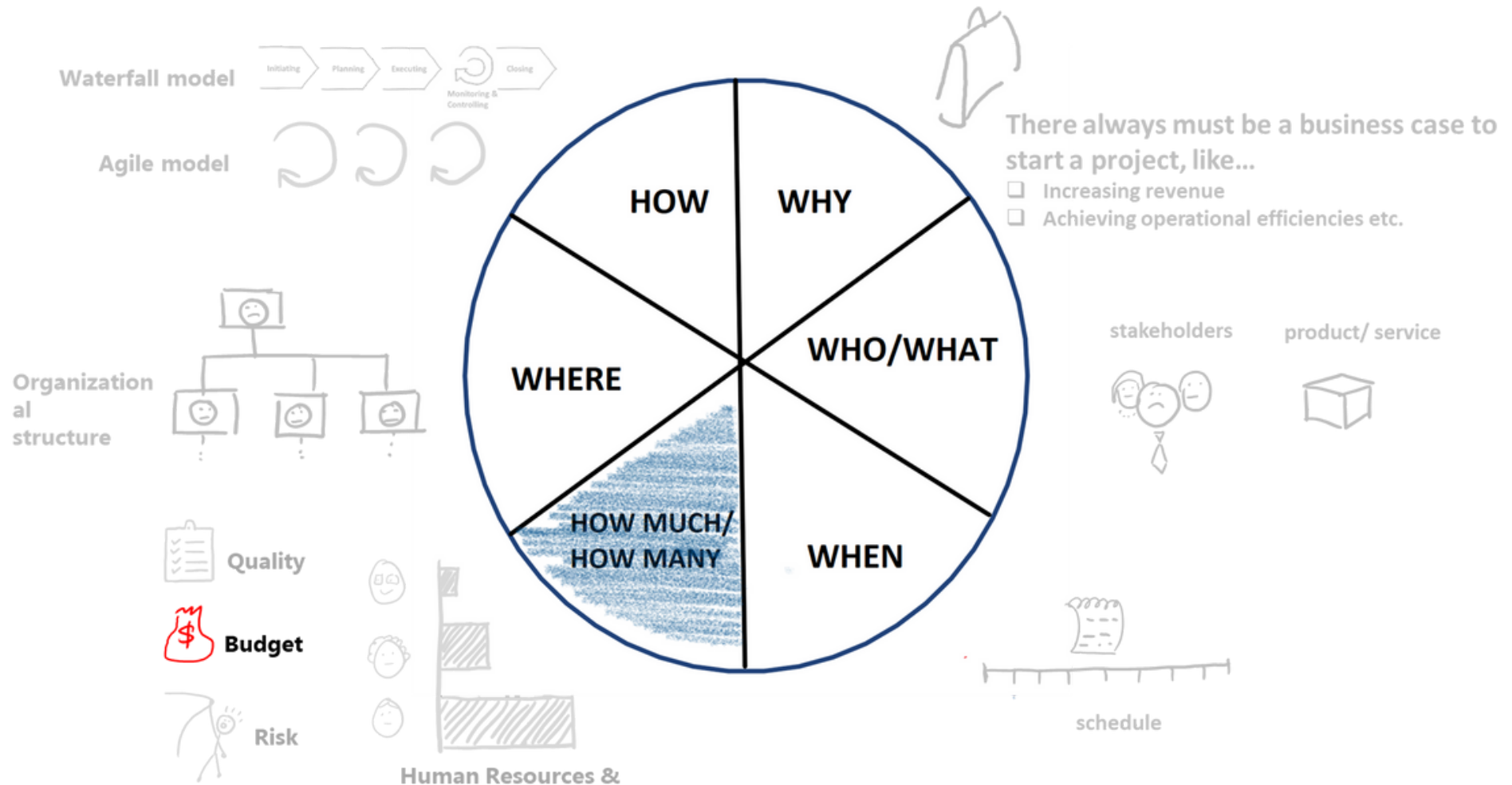
- Schedule Network Analysis
- Critical Path Method
- Leads and lags
- Schedule compression, etc.



Project Schedule

-
- Three main paths in the diagram are: preparing stone blocks, preparing mortar, and preparing surroundings
 - Critical path method determines the longest chain of tasks, which is the path for preparing stone blocks, cutting them, transporting, and building inside passages and final chamber, taking 18 months
 - Activities on the critical path cannot be delayed and have zero float
 - Other activities have float and can be delayed without jeopardizing the project schedule
 - Final project schedule includes key milestones, including the date for final product delivery.

HOW MUCH/MANY DO WE NEED



-
- ❑ The project schedule is an important deliverable that shows when things will happen on the project.
 - ❑ The cost, quality, resource, and risk management knowledge areas belong to the "how much" and "how many" portion of the project.
 - ❑ Cost management involves estimating the amount of money needed to perform each activity based on past projects or current estimates.

HOW MUCH/MANY DO WE NEED

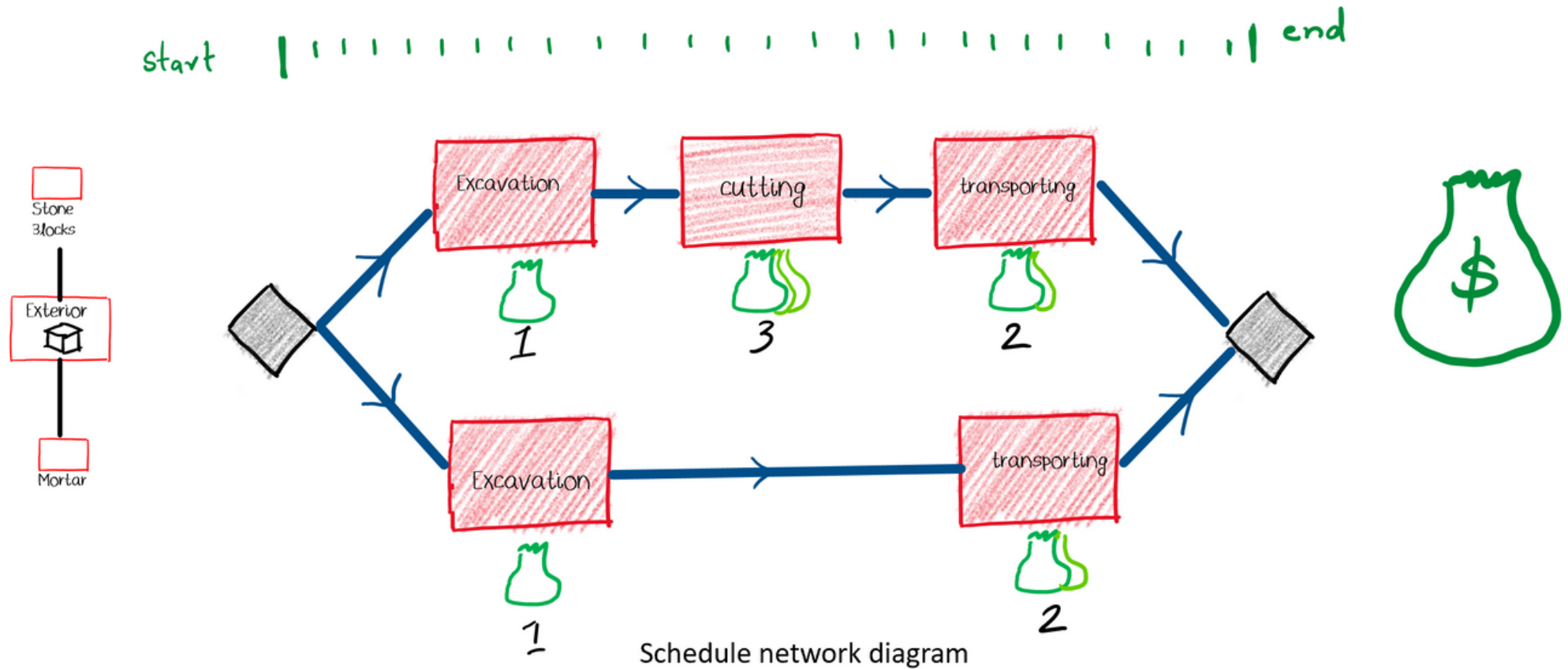
HOW MUCH/MANY DO WE NEED



-
- The estimate costs process produces activity cost estimates that are associated with specific points in time on the project schedule.
 - Estimating costs is a process within the cost management knowledge area of project management that involves estimating the amount of money required to complete the project activities.
 - The process of estimating costs typically involves breaking down the project scope into individual activities and estimating the cost of each activity based on the resources required, such as labor, materials, and equipment.

WHEN ARE THINGS TO BE DONE

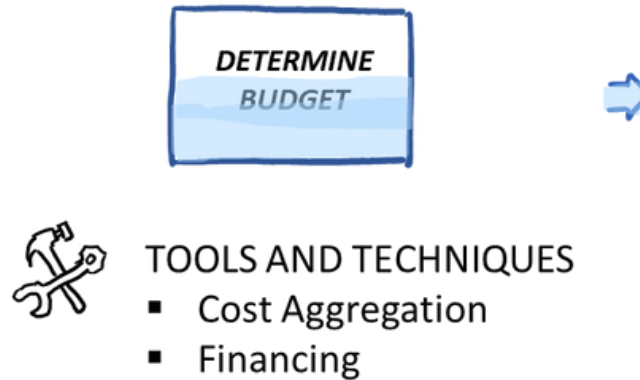
ESTIMATING ACTIVITY DURATIONS



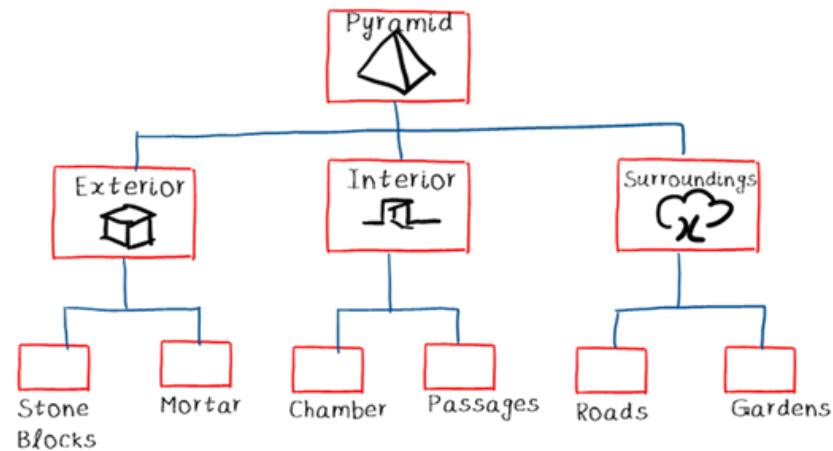
Cost estimates can be associated with work packages on the project, such as building the stone blocks or working on the mortar.

HOW MUCH/MANY DO WE NEED

COST MANAGEMENT



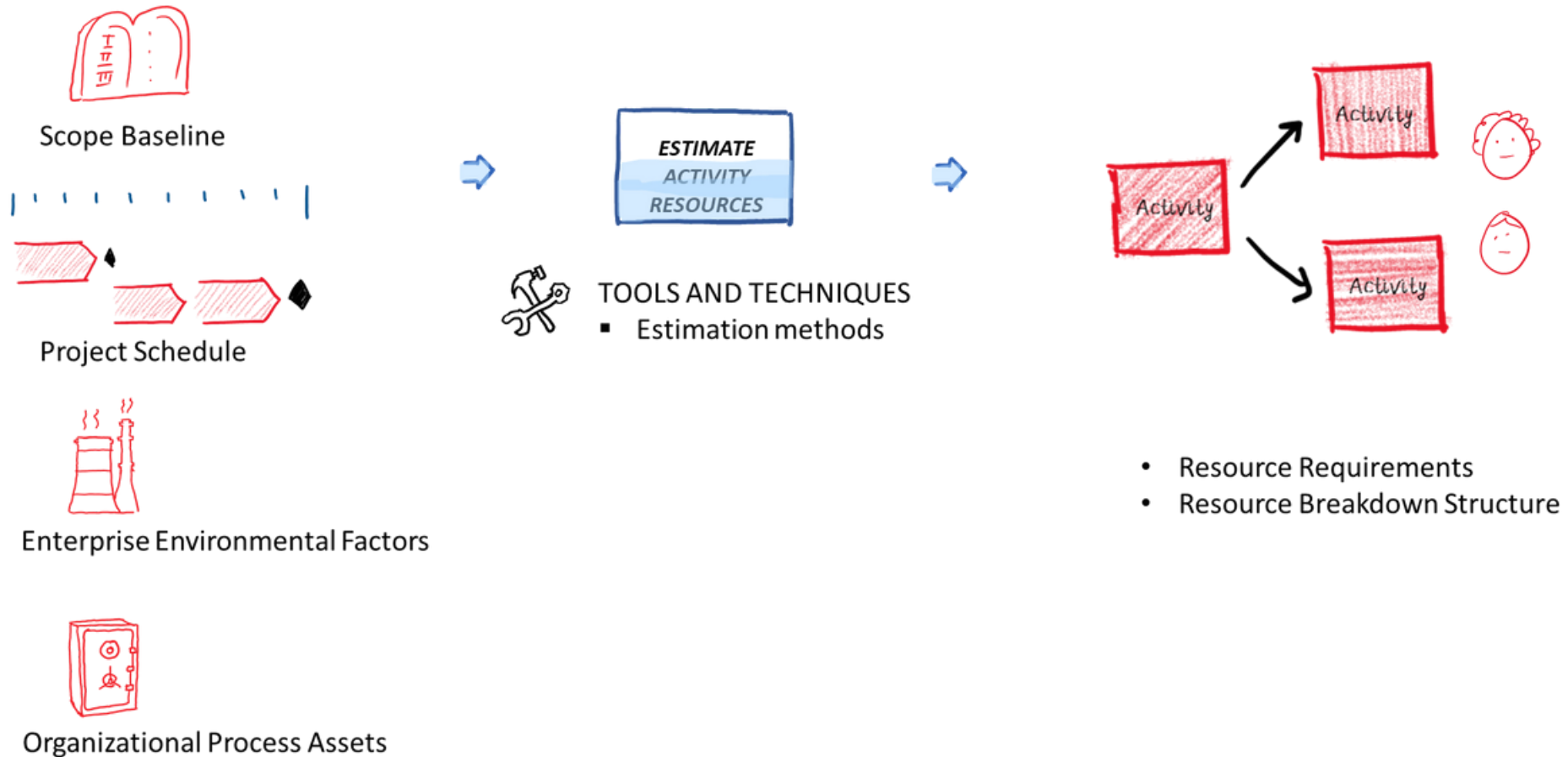
Project Budget



-
- The final process in the cost management knowledge area is Determine Budget process
 - This process aggregates the cost from each activity estimated in the previous process
 - Aggregation can be done to higher levels on the WBS structure, called control accounts, for easier monitoring and control of project budget on smaller components
 - Control accounts can be broken down by various departmental activities or other scope of the project
 - The project budget is an important deliverable from the cost management knowledge area that determines how much money will be spent on the project

HOW MUCH/MANY DO WE NEED

RESOURCE MANAGEMENT



-
- The resource management knowledge area focuses on estimating the number of resources needed for each project activity.
 - The first process in this knowledge area is "estimate activity resources", which uses inputs from previous processes to determine the number of resources needed for each activity.
 - The output of this process includes the resource requirements document and the resource breakdown structure.
 - Sometimes, determining the resource assignments on each activity is necessary to estimate the costs that will be incurred on each activity.
 - The resource assignments on each activity are illustrated in the resource breakdown structure.
 - Through this knowledge area, the project team quantifies the amount of money needed to execute the project and the number of resources with relevant skills needed to work on the project.

HOW MUCH/MANY DO WE NEED

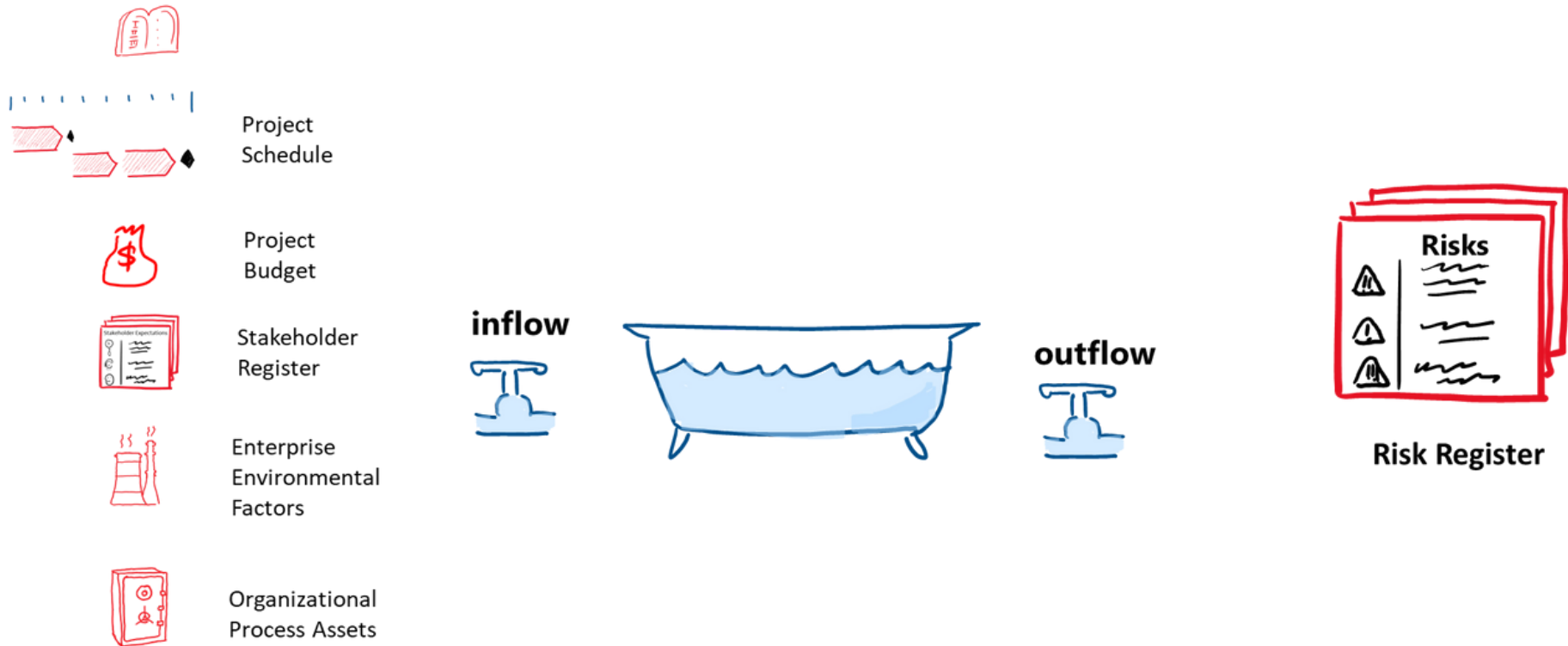
QUALITY MANAGEMENT



-
- ❑ Delivering a project within scope, budget, and time is necessary but not sufficient for success.
 - ❑ Quality is the measure of satisfying customer or sponsor requirements for the product or service.
 - ❑ There is product quality (satisfying customer needs) and project quality (efficient execution).
 - ❑ Quality metrics can include completed tasks, schedule delays, cost overruns, defects, and stakeholder expectations.
 - ❑ Plan quality process involves input from all three baselines (scope, schedule, cost) and the stakeholder register.
 - ❑ Quality metrics are important for project communication and overall project status.
 - ❑ Testing is used to determine product quality, while benchmarking and testing are used to determine project quality.
 - ❑ Quality improvements can be derived from comparing previous similar activities to the current project activities.

HOW MUCH/MANY DO WE NEED

RISK MANAGEMENT



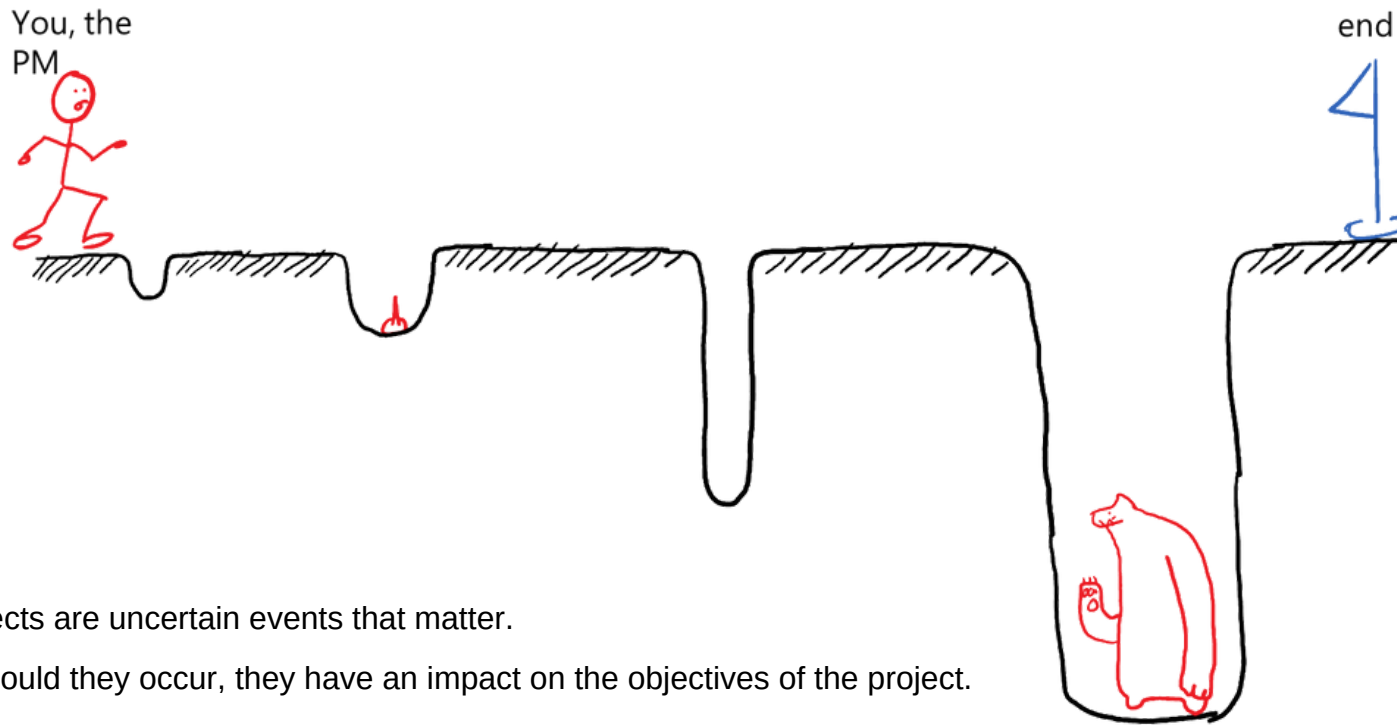
RISKS on projects are uncertain events that matter.

This means, should they occur, they have an impact on the objectives of the project.

-
- ❑ Risk management is a critical aspect of project management that involves identifying, analyzing, and mitigating potential risks that could impact the success of a project.
 - ❑ Projects operate in uncertainties and risks are uncertain events that matter to the project and could affect its objectives.
 - ❑ Not all uncertainties are risks to the project, only those that matter should be considered.
 - ❑ Risks present varying degrees of danger and should be navigated to reach successful completion of the project.
 - ❑ Risk management in project management aims to minimize the likelihood and impact of negative events or risks, while maximizing the chances of positive outcomes.

HOW MUCH/MANY DO WE NEED

RISK MANAGEMENT



RISKS on projects are uncertain events that matter.

This means, should they occur, they have an impact on the objectives of the project.

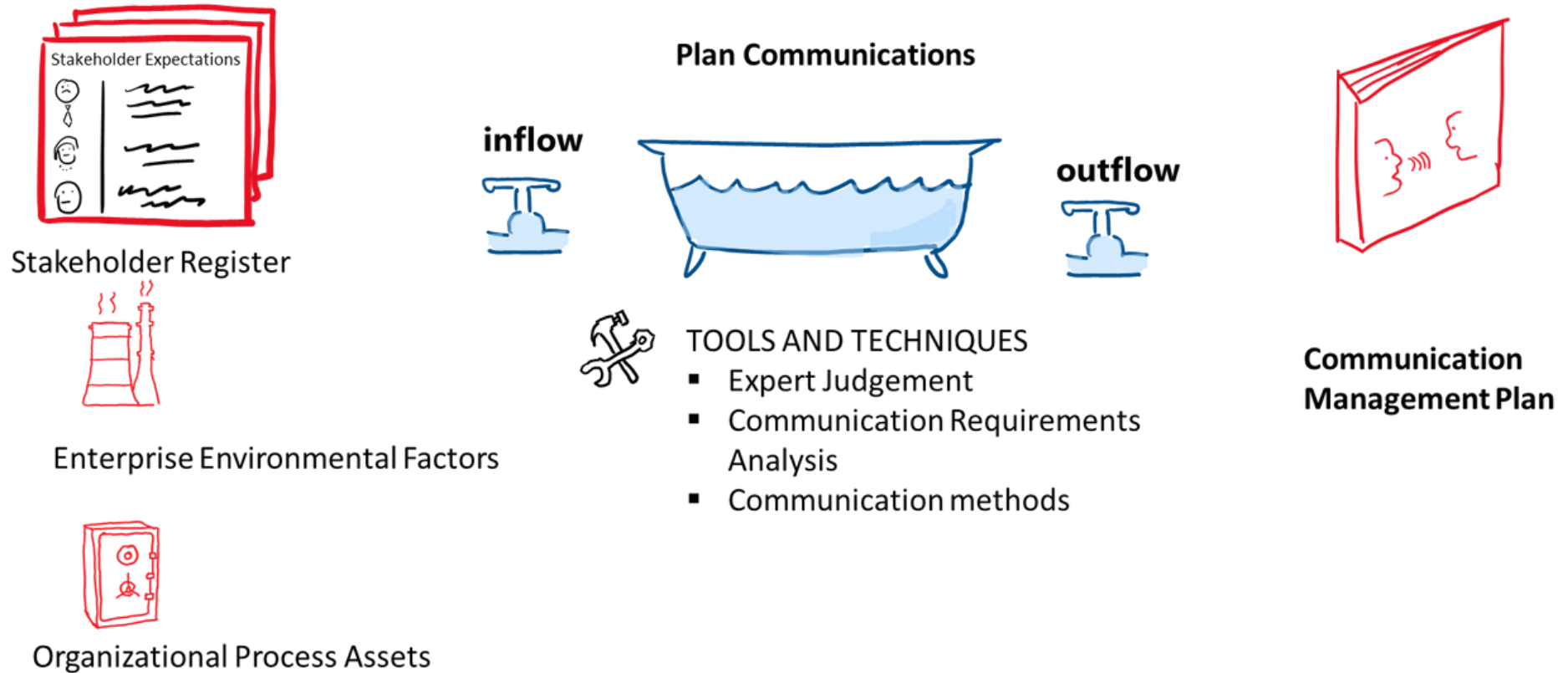
- This illustration discusses the importance of risk management in project management.
- Risks are uncertain events that matter to the project and can affect its objectives.
- The illustration is about navigating a landscape with holes of varying degrees of danger to explain the concept of risks.
- The size of the hole and the potential harm it poses are used as dimensions to quantify and measure risks for analysis.
- The emphasis here is that not all uncertainties are risks, only those that matter to the project should be considered.

HOW MUCH/MANY DO WE NEED



-
- ❑ Risk management planning involves three passes to create the risk register.
 - ❑ The risk register is the output of each process and gets updated every time.
 - ❑ The first process is to identify risks by looking at information about the project, including scope, schedule, budget, quality, human resources, and external factors.
 - ❑ Risks are documented in the risk register, including the risk event, its source, its impact, and the risk owner.
 - ❑ Tools and techniques used to identify risks include expert judgment, data gathering and analysis, meetings, and SWOT analysis.
 - ❑ The risk register is updated with additional information at each step of the process.

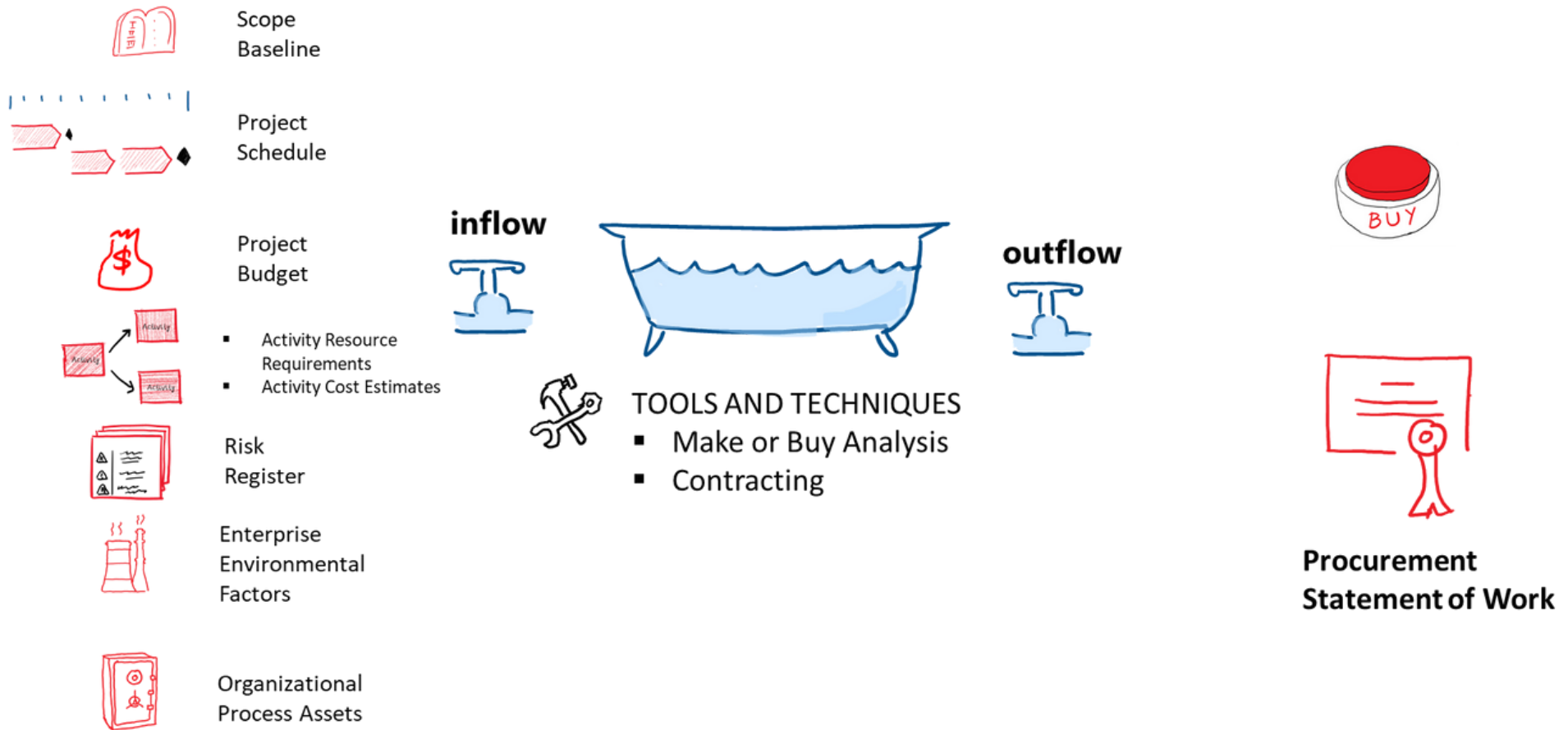
WHERE DO WE REACH OUT



-
- During an ongoing project, project communication is the most important output for customers or sponsors.
 - Communication from the project manager is key to keeping stakeholders happy, motivated, and supportive of the project.
 - Knowing where to communicate within and outside the organization is important, and different stakeholders require different levels of information.
 - Communication requirements are documented in the communication plan, which outlines what stakeholders need to know, when they need to know it, and in what format.
 - The communication plan is an important output that includes information on where and how to communicate with relevant stakeholders, and it is informed by the stakeholder register.

WHERE DO WE REACH OUT

WHERE DO WE REACH OUT



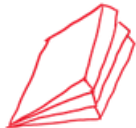
-
- Procurement management in project management is the process of identifying, acquiring, and managing external goods and services needed to complete a project.
 - It involves selecting the right vendors or contractors, negotiating contracts, and ensuring that the work is completed on time, within budget, and meets the project's quality requirements
 - The process involves identifying what is needed, how much, and when it will be available.
 - Inputs to Plan Procurement Process includes documents related to the project's scope, activities, budget, and timelines.
 - Outputs include make or buy decisions, a procurement statement of work, and contract types.
 - Make or buy analysis is used to determine whether a contractor or agency needs to be engaged.
- The procurement statement of work includes conditions of work, scope, timelines, and cost.

HOW TO DO THE PROJECT INTEGRATION MANAGEMENT



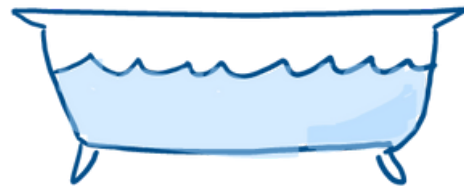
Project
Charter

Develop Project Management Plan



- ☐ Scope Mgmt. Plan
- ☐ Cost Mgmt. Plan
- ☐ Schedule Mgmt. Plan
- ☐ Quality Mgmt. Plan
- ☐ Resource Mgmt. Plan
- ☐ Risk Mgmt. Plan
- ☐ Communication Mgmt. Plan
- ☐ Procurement Mgmt. Plan

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**Project
Management
Plan**



Enterprise
Environmental
Factors



TOOLS AND TECHNIQUES

- Expert Judgement

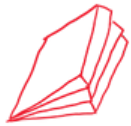


Organizational
Process Assets

-
- ❑ "Develop project management plan" is the final process in the planning phase.
 - ❑ The output of this process is the project management plan.
 - ❑ The project management plan outlines how the project will be done in subsequent phases and provides a roadmap to completion within the constraints of scope, schedule, and budget.
 - ❑ The inputs to this process are the project charter and individual management plans for scope, cost, schedule, quality, resources, risk, communications, and procurement.
 - ❑ These inputs are reviewed and consolidated to create the project management plan.

HOW TO DO THE PROJECT

INTEGRATION MANAGEMENT



All management plans



Project communications



Risk Register



Requirements Traceability Matrix



Milestone List & Schedule



Enterprise Environmental Factors



Organizational Process Assets

Direct and Manage Project Work

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outflow



TOOLS AND TECHNIQUES

- Expert Judgement
- Project Management Information System
- Meetings



Product/ Service Deliverables

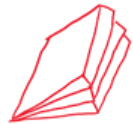


Work Performance Information

-
- The executing phase of the project is where the real action takes place in terms of building the product or service.
 - The approved project management plan is used as a roadmap for the project, and resource allocation is in place.
 - The process used to guide the executing phase is called direct and manage project work in the integration management knowledge area.
 - Deliverables are created during the executing phase, and work performance information is collected to track how the project is performing.
 - Project management information systems and other tools are used to manage the project and track data.
 - Inputs to the executing phase include management plans, risk register, scope baseline, requirements traceability matrix, milestone lists, schedule, project communications, and stakeholder engagement.
 - During the executing phase, quality, resources, teams, and communications are managed, and procurement is conducted to onboard external vendors to perform the job on the project.

HOW TO DO THE PROJECT

INTEGRATION MANAGEMENT



All management plans



Deliverables



Stakeholder Register



Enterprise
Environmental
Factors



Organizational
Process Assets

Manage Project Knowledge



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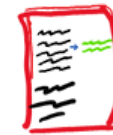


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TOOLS AND TECHNIQUES

- Expert Judgement
- Knowledge and information management



Lessons Learned
Register

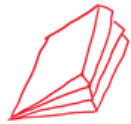


Organizational
Process Assets
updates

-
- Lessons Learned are important observations and insights obtained from a project
 - It is important to keep a record of Lessons Learned for the benefit of future projects and the organization
 - Manage Project Knowledge is a process in the executing phase that sits in the integration management knowledge area
 - Inputs include project management plans, various deliverables, and the stakeholder register
 - The output is the Lessons Learned register, which tracks all the observations and insights obtained from the project
 - The Lessons Learned register is updated and stored in the organizational assets for later use.

HOW TO DO THE PROJECT

INTEGRATION MANAGEMENT



All management plans



Risk Register



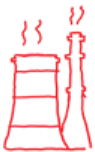
Quality Metrics



Work Performance
Information



Schedule



Enterprise
Environmental
Factors



Organizational
Process Assets

Monitor and Control Project Work

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TOOLS AND TECHNIQUES

- Expert Judgement
- Data Analysis
- Meetings



**Change
Requests**

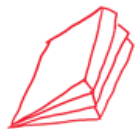


**Work
Performance
Reports**

-
- The Project Manager monitors and controls the project work during the execution phase.
 - The process for this is called "Monitor and Control project work".
 - The Project Manager ensures that the project work is on track and checks for deviations from the management plans.
 - Data collected during the executing phase is used to create work performance reports, which are used to communicate project progress to stakeholders.
 - Change requests may arise during this phase, such as requests for additional scope or time extension.
 - Inputs for this process include project management plans, risk register, and quality metrics.

HOW TO DO THE PROJECT

INTEGRATION MANAGEMENT



- ☐ Scope Mgmt. Plan
- ☐ Cost Mgmt. Plan
- ☐ Schedule Mgmt. Plan



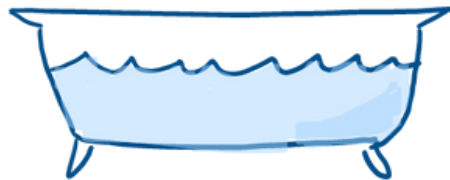
Enterprise
Environmental
Factors



Organizational
Process Assets

Perform Integrated Change Control

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outflow



TOOLS AND TECHNIQUES

- Expert Judgement
- Change control tools
- Meetings

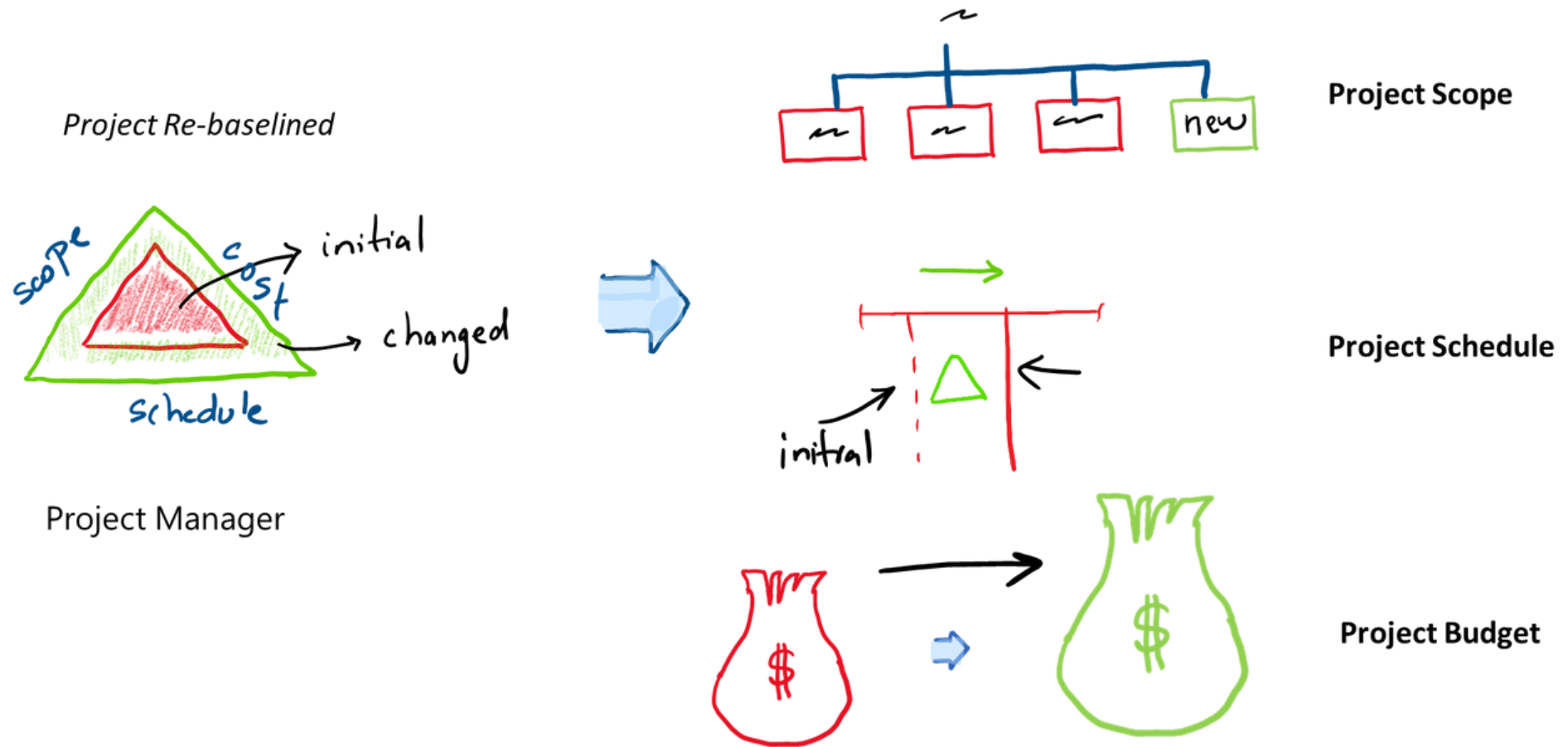


**Approved
Change
Requests**

-
- Changes to a project can be introduced in the middle of the project that need to be managed through a systematic process
 - The baseline documents encapsulate the 100% scope of the project agreed upon by stakeholders
 - A change request is a formal proposal to modify any project document or baseline
 - The perform integrated change control process manages changes introduced to the project
 - Inputs to this process include all project management plans prepared so far, such as the scope, schedule, and cost management plans
 - The output of this process is approved change requests
 - Tools and techniques used in this process include expert judgment, change control tools, and meetings
 - Controlling a project to the baselines is an essential part of project management
 - The project manager is responsible for executing the change management process for each requested change.

HOW TO DO THE PROJECT

INTEGRATED CHANGE CONTROL



-
- Changes to a project can be introduced in the middle of the project that need to be managed through a systematic process
 - The baseline documents encapsulate the 100% scope of the project agreed upon by stakeholders
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HOW MUCH/MANY DO WE NEED

COST MANAGEMENT

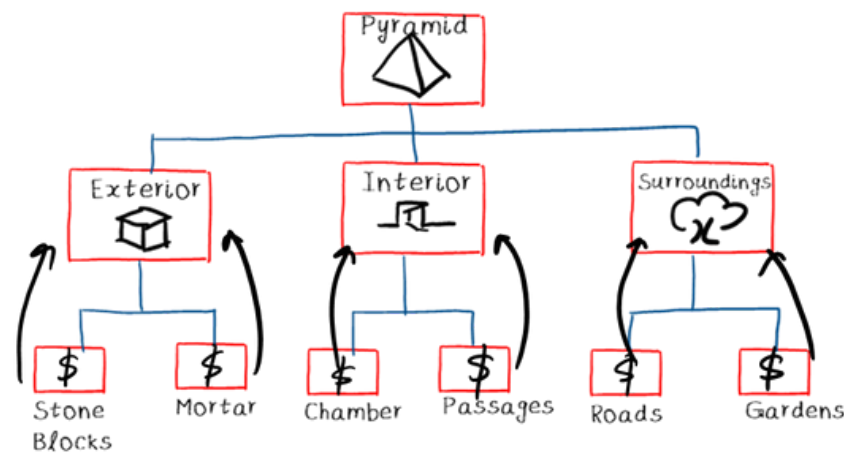


Project Budget



TOOLS AND TECHNIQUES

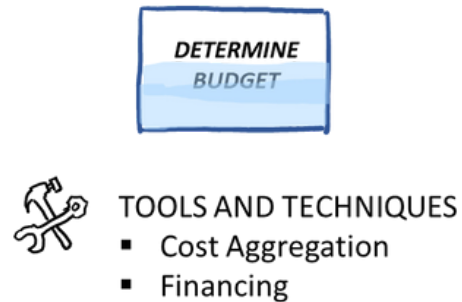
- Cost Aggregation
- Financing



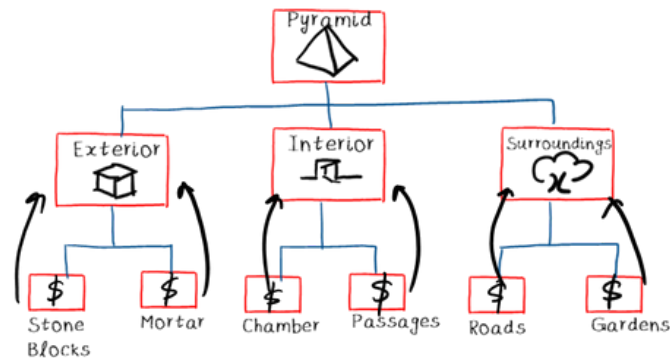
-
- Project budget is obtained through cost estimation methods and rolling up amounts from control accounts.
 - Project manager's job is to ensure project is delivered within the defined budget.
 - Control Costs is the process used to monitor and control project costs.
 - Earned Value Management is an important concept in project management.
 - Inputs to the Control Costs process include project budget, project management plan, and work performance information.
 - Outputs of the process include budget forecasts and metrics related to costs and budget.
 - Tools and techniques used include Earned Value Management and project management software.

HOW MUCH/MANY DO WE NEED

COST MANAGEMENT

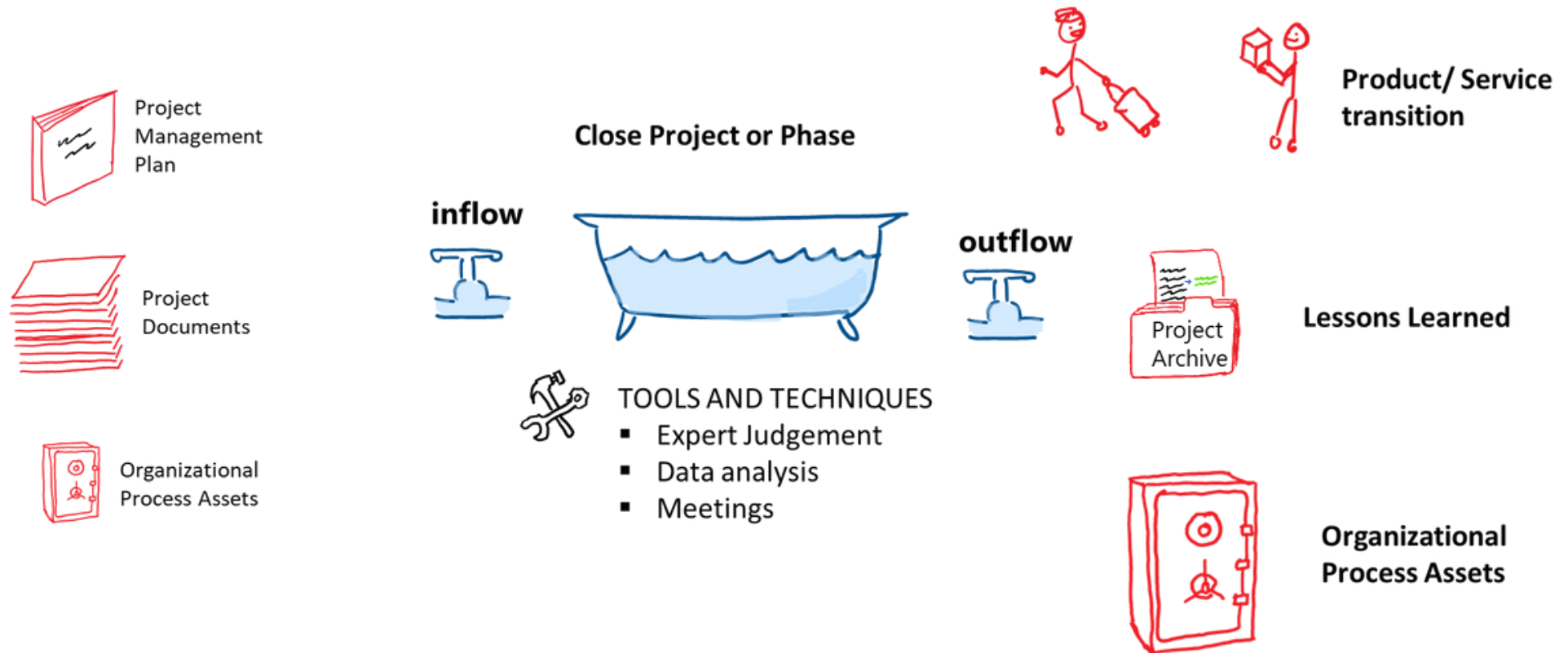


Project Budget



- Project budget includes cost estimates, contingency reserves for risk responses, and management reserves for unplanned changes.
- The overall project budget is called the budget at completion (BAC) and includes the initial project budget, contingency reserves, and management reserves.

HOW TO DO THE PROJECT INTEGRATION MANAGEMENT



-
- The closing phase is the final phase of the project, leading to its conclusion.
 - The project team delivers the product or service and obtains final acceptance from the sponsor or customer.
 - If the project delivers a service, it may be transitioned to a separate team for maintenance and support.
 - The closing phase involves the process of Closed Project or Phase.
 - Inputs to this process include the project management plan, project documents, and organizational process assets.
 - Outputs of the process include updating lessons learned, gathering insights from stakeholders, and identifying areas for improvement.
 - Lessons learned help identify areas for improvement and guide future projects.
- The project manager archives all project documents into organizational process assets for future reference and guidance.



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